Supplementary Material

State Aggregation in Dynamic Graph Recovery:
Separating Structural Ranking from Temporal Discrimination

Xuansong Chang

October 2026

Evidence guide. Sections S1–S3 define the targets, procedures and diagnostic identities. Section S4 reports Studies A–C: A's pre-specified four-contrast family uses 100 new paired repeats; B's separately specified schedule comparisons use 50 different repeats; C is a separately specified robustness study with 30 new repeats. Section S5 separates additional controls on existing outputs from the post hoc AP/threshold and residual-pair sensitivity analysis. Section S6 retains the earlier exploratory grid (seeds 0–19), a separate auxiliary independent validation (seeds 3000–3049), representation probes and a fixed segmentation/refit comparison on existing C inputs. These blocks retain their own statistical units and comparison families. Section S7 treats static industrial alignment as a separate reference target. Section S8 distinguishes cached-output replay from selected same-seed reruns and locates the implementation records.

S1 Evaluation targets and notation

Let $X(1{:}T) \in \mathbb{R}^{T\times N}$ be observed, with latent dynamic state $z(t) \in \{1,\dots,K_{\mathrm{dyn}}\}$ and directed lag-specific support $A[k,\ell,i,j] = 1$ when $i(t-\ell)$ directly precedes $j(t)$ in state k. For window length h and stride s, the window count is $M = 1 + \lfloor (T-h)/s \rfloor$, with indices $w = 1,\dots,M$. Earlier formulas in this supplement use W for the same scalar window count; there $W = M$. The main article uses M throughout.

Union graph. $A_{\cup}[\ell,i,j] = 1\{\exists k : A[k,\ell,i,j] = 1\}$, scored by AUROC, AUPRC, precision at the true union edge count, and SHD.

Regime-specific graphs. For each state k, the estimated $G[k]$ is compared with $A[k]$ after matching state identities by maximum overlap. We report macro AUPRC and AUROC over matched states with non-empty support, the prevalence of regime edges, the AUPRC lift over prevalence, and the normalized AP $(\mathrm{AUPRC} - \pi)/(1-\pi)$.

Temporal state sequence. Window assignments are compared with the true regime at the window centre, with ARI, macro F1, the signed state-count error $\hat{K} - K_{\mathrm{dyn}}$, and state-label boundary F1 at a one-window tolerance. This earlier label-sequence metric is distinct from the score-change slot target in Section S5.2.

Three state counts. K_{true} is the number of dynamical states in the generator, $K_{\text{candidate}}$ is the count handed to a method, and $\hat{K}$ is what the method returns. Track I supplies $K_{\text{candidate}} = K_{\text{true}}$. A *gate-stress* condition supplies $K_{\text{candidate}} = 2$ when $K_{\text{true}} = 1$ on purpose, to measure false splits. Track II would supply no count; we define it but do not claim it as an empirical contribution.

Two structural counts. K_{dyn} counts distinct data-generating mechanisms; K_{graph} counts distinct binary supports. They coincide for a support change or a lag shift, but not for the mechanism and nuisance families: a sign flip, resampled magnitudes, a variance change, a mean shift or a state-specific trend produce $K_{\text{dyn}} = 2$ with $K_{\text{graph}} = 1$. When $K_{\text{graph}} = 1$ the regime-specific binary metric – and likewise the pooled time-indexed binary target, whose ground truth A_{z_t} is then the constant graph A – still scores support recovery but cannot tell whether the method distinguished the dynamical states; that question is measured by state-label alignment (ARI) alone. Weighted or signed mechanisms are a third target that none of the binary metrics evaluates; we report it as a limitation rather than as a result.

Time-indexed graph scores. For an evaluated nonself lagged edge e, let $A_{w,e}$ be its support at the integer centre of window w and $S_{w,e}$ its emitted score. The primary within-edge temporal AUROC, $\mathrm{T-AUC}_{\Delta}$, is the macro average of $\mathrm{AUROC}((A_{w,e})_w, (S_{w,e})_w)$ over edges whose support takes both values in the evaluation windows. Both positive and negative windows are required; an undefined target remains missing, while constant scores on an evaluable edge score 1/2. Pooled AUROC instead ranks all window–edge entries together and mixes within-edge and cross-edge comparisons (Section S3.2). Neither score measures boundary distance.

Distinct recovery questions. Boundary detection uses intervals between consecutive window centres. A slot is positive when an edge's true support changes at least once in that interval; the predicted change strength is the absolute difference between consecutive emitted scores. This is change detection at stride resolution. Section S5.2 defines coverage, AP and threshold conventions.

Binary support is evaluated through union, regime-specific and time-indexed targets. State assignments are assessed against the centre labels by ARI and related label metrics; J in Section S3.2 specifically denotes state-assignment discrimination. Score-change boundary detection is defined separately in Section S5.2. Signed and weighted mechanism recovery is not evaluated: the pipeline scores absolute coefficients. When all states share one binary support, graph scores assess that support without identifying whether the dynamical states were distinguished.

S2 Generators, local scorers, and comparison procedures

S2.1 Studies A and B: VAR generator and core defaults

Studies A and B share the VAR generator specified below: N=8 nodes, L=3 lags, base edge probability 0.18, effect scale 1.0, Gaussian innovations with scale 0.10, no observation noise and 100 burn-in samples. Their record lengths, duration and schedule conditions vary as specified in Section S4.1. Table S1 collects the reference scorer and matching settings. Study C uses the separate symmetric construction in Section S4.2.

Coefficient, support and phase construction. For each lag, state-A nonself supports are independent Bernoulli draws with probability $p = 0.18$, and all diagonal positions are set to zero. If the entire tensor is empty, a source i is sampled uniformly from $\{0, \ldots, N-1\}$ and the lag-one

edge $i \to (i+1) \bmod N$ is inserted. For the support-changing family, state B retains each A edge independently with probability o (the configured overlap), then adds each position outside the *original* A support with probability $p(1-o)$. The addition mask is not recomputed after deleting A edges. Ignoring both empty-support safeguards, the marginal nonself-edge probability in B is $po + (1-p)p(1-o) = 0.1638$ at $p = 0.18, o = 0.5$, rather than 0.18. If the resulting entire B tensor is empty, it is replaced by the full A support. The addition mask spans the full tensor, including the initially absent diagonal slots, so state B can contain self-lag terms. Generated self-lags enter the VAR recursion and regression design; evaluation and clustering use nonself entries only. State A has no self terms while state B may have them. These asymmetric support properties are held fixed in Studies A and B; Study C evaluates a separate symmetric construction. Before the fallback, each B diagonal position has probability $p(1-o) = 0.09$. Thus o is a retention probability, not a guaranteed realised overlap or Jaccard index. Realised overlap is the shared edge count divided by the state-A edge count. Each supported coefficient at lag ℓ has an independent equiprobable sign and magnitude $\text{Uniform}(0.08, 0.35)\, a/\ell$, where a is effect scale. Coefficients are resampled in state B, including shared edges, except in the specified identical-coefficient nuisance controls. The sign-flip control uses the negative of the first-state tensor.

In the two-state branch, a tensor whose initial companion radius ρ exceeds 0.90 has every lag block multiplied once by $0.90/\rho$; this is the executed rescaling rule. For $L > 1$ it does not multiply the entire companion matrix by $0.90/\rho$, and need not make its final radius equal to 0.90. It is not a joint-switching stability guarantee. The one-state auxiliary branch returns its sampled tensor directly. Path stability is assessed separately in Section S6.1. The first L simulation rows are independent $\mathcal{N}(0, 0.10^2)$ values and later rows follow the VAR recursion with fresh innovations. The fixed schedule starts in state A, alternates after D steps, and is generated for $T + 100$ steps before discarding 100 burn-in steps. Consequently observed time zero starts 100 steps into the original schedule. This phase convention applies to the fixed-schedule reference; Study B's random-phase and irregular schedules are specified in Section S4.1.

Earlier reference configurations. The four earlier auxiliary scenarios use T=1200, window length h=300, stride 150 and $K_{\text{candidate}}$=2, with the generator defaults above. In the switching reference, each state lasts D=300 steps, the full cycle is P=600 and $h = D$. The 100-step burn-in is not a multiple of D, so the observed switch points fall off the window grid. Scans and control conditions override these reference settings as specified below.

The four auxiliary scenarios differ only as follows: *switching core* (`overlap`=0.5), *state-specific* (`overlap`=0.0), *shared graph* (`control=same_support`, `overlap`=1.0, second state resampled on the same support), and *stationary* (`control=stationary`, one state). Table S25 displays the four configured scenarios using twenty exploratory seeds per scenario; Table S16 shows the earlier switching/stationary panels. The auxiliary independent validation uses fifty different seeds paired across these same scenario definitions and reports contrasts after averaging the four scenarios within each seed (Table S28, Section S6.8). The exploratory and validation tables therefore describe distinct samples. The archive retains the full configuration fields.

S2.2 Local scores and hard-emission policies

We compare emission policies on shared local graph scores. Gated hard uses a static window-maximum output on rejection. For each window we fit one Lasso per target on lagged standardised inputs and take absolute coefficients as the window score. Self-lags enter the regression design but are excluded from the scored edges and clustering features. Features are the flattened non-self scores, standardised across windows; states come from K-means with a supplied count. State

Setting	Value
Generator	switching VAR(3), N=8, burn-in 100; rescaling rule in text
Switch schedule	D=300 per state (P=600); burn-in 100 offsets the window grid
Coefficient sampling	independent sign ± 1, magnitude $U(0.08, 0.35)/\ell$ at lag ℓ
Innovation noise	Gaussian, scale 0.10; observation noise 0.0
Scoring	per-variable z-score, then Lasso per target on the lagged design
Lasso	$\alpha = 0.005$ (fixed, not tuned), `fit_intercept=False`, `max_iter=10000`
Design	lag-major: all $L \times N$ lagged predictors, both self- and cross-lags
Self-lags	fitted in the regression, excluded from edge score and clustering features
Standardization	per-variable z-score; design and target z-scored again before Lasso
Clustering features	flattened non-self absolute scores, standardized across windows
State assignment	K-means, $K_{\text{candidate}}$=2 (gate per Section S2.2)
State matching	Hungarian assignment by maximum window overlap (Section S1)
Window	h=300, stride 150

Table S1: Core simulation defaults for the generator, scorer and state matching; scans override the stated settings, and ESS has a separate protocol. "Self-lags" distinguishes the regression input (self-lags enter the design) from the edge score and clustering features (self-lag scores are zeroed).

scores are the within-state mean of window scores, and the union score is the maximum over state scores. For time-indexed evaluation, an accepted partition emits its assigned group map at each window. If the gate rejects, gated hard emits the same window-maximum map at every window. The gate accepts when the K-means improvement is at least 0.12, or when it is at least 0.04 with temporal coherence at least 0.55 and a minimum cluster fraction at least 0.10. Improvement is $(I_1 - I_K)/I_1$, clipped below at zero, where I_1 and I_K are the one-group and fitted K-means inertias of the standardised window features; it is set to zero when $I_1 \leq 10^{-8}$. For the candidate labels before this decision, define empirical group fractions $f_k = M^{-1} \sum_w \mathbf{1}(\widehat{z}_w = k)$, adjacent agreement $u = (M-1)^{-1} \sum_{w=2}^{M} \mathbf{1}(\widehat{z}_w = \widehat{z}_{w-1})$ and $q = \sum_k f_k^2$. The executed coherence is $\max\{0, (u-q)/(1-q)\}$. It is zero for fewer than two windows, a single candidate group or $q \geq 1 - 10^{-8}$. No true labels enter this calculation.

Forced hard emits the same fitted candidate maps and assignments regardless of the gate. Raw fallback uses forced hard on acceptance and each window's raw scores on rejection; its acceptance decision is unchanged. Raw window and static window mean are separate controls. Oracle-label aggregation averages by true window labels, without refitting. In the earlier seven-window core all gates accept, so the gated and forced hard outputs coincide there; the distinction is essential when acceptance is incomplete.

S2.3 Partitioning and legal contiguous-range refitting

The PELT baseline partitions the same window features (five principal components, penalty $d_{\text{eff}} \log W$, jump = 1). A partition can be scored by *window averaging* (average existing window scores within each group) or by *refit* (re-estimate Lasso on each group's contiguous time ranges, with boundaries at window-centre midpoints). A group that recurs in several runs is refit *per run* and the runs are then averaged, so no artificial lag links are created across discontinuities; the runs of one state are not pooled into a single design matrix. Crossing the two partitions with the two rules gives the 2×2 factorial of Section S6.6. A global VAR-Lasso with no partition is reported as a reference.

At seven windows the BIC-scaled penalty $d_{\text{eff}} \log W$ is too small to merge any pair of windows,

so PELT returns one segment per window; a brute-force check over all $2^6 = 64$ contiguous partitions confirms that this is the global optimum of the penalised cost (Section S6.7). The penalty scan shows the realised count is set by the minimum segment length, not by the penalty factor.

Because a penalised partition and a supplied two-state partition are not the same estimator, we add three comparators with the segment count held fixed at two: the best two contiguous segments, the best split of a raw-VAR Chow scan, and a split at the first true change point. This separates "a better change-point detector" from "modelling recurring regimes".

S2.4 Known, candidate, and returned state counts

Table S2 makes the protocol explicit. Supplying the true count on a stationary series yields a one-state answer ($\hat{K} = 1$, ARI 1.000, gate rejection rate 1.00). Supplying the wrong candidate ($K_{\text{candidate}} = 2$) produces a two-state answer with ARI 0.000 and raises union AUROC from 0.893 to 0.908: the false split is not harmful on the union metric.

Table S2: Protocol check. The gate-stress row is deliberately given the wrong candidate count.

Condition	K_{true}	K_{cand}	$\hat{K}$	Accepted	Union AUROC	Assignment ARI
Stationary, Gate stress	1	2	2.0	1.00	0.908	+0.000
Stationary, Track I	1	1	1.0	0.00	0.893	+1.000
Switching, Track I	2	2	2.0	1.00	0.801	+0.292
Switching, Under-specified	2	1	1.0	0.00	0.804	+0.000

S3 Derivations and diagnostic identities

The pipeline links the observed series to local representations, inferred groups, state maps and evaluation targets:

$$\text{raw series} \to \text{window representation} \to \text{state estimate} \to \text{state graph} \to \text{evaluation}.$$

The analyses below have distinct evidence roles. Gate geometry, occupancy ranking and the two-state factorisation are mathematical statements under their stated conditions; their numerical reconstructions check implementation consistency. Window-mixture calculations and the fixed-mechanism denoising intervention probe representation and estimation. The Brier analysis concerns probabilistic risk, rather than explaining the Lasso/AUROC ranking differences.

S3.1 Static occupancy ranking and its diagnostic ceiling

For a time-constant score s_e per edge and per-edge activation frequency $\pi_e = \frac{1}{W}\sum_w A_{w,e}$, writing $H(u) = \mathbf{1}(u > 0) + \frac{1}{2}\mathbf{1}(u = 0)$, the pooled AUROC over all "window $\times$ edge" pairs is exactly

$$\text{AUC}_{\text{static}} = \frac{\sum_{e,f} \pi_e(1-\pi_f)\, H(s_e - s_f)}{\left(\sum_e \pi_e\right)\left(\sum_f (1-\pi_f)\right)}. \tag{S1}$$

Only the activation *frequencies* π_e appear, never the times at which each edge activates. Permuting each edge's activation times (holding counts fixed) therefore leaves a constant prediction's pooled AUROC unchanged, and the optimal static ordering ranks edges by π_e. A static "occupancy oracle"

that knows which edges are more frequent but not when they occur is thus the natural diagnostic ceiling.

Maximising Eq. (S1) over orderings gives a closed-form ceiling. Writing $P = \sum_e \pi_e$ and $Q = \sum_e (1 - \pi_e)$, ranking by π_e is optimal (swapping two edges changes the numerator by $\pi_e(1-\pi_f) - \pi_f(1-\pi_e) = \pi_e - \pi_f$, non-negative under the sorted order), and the pairwise contributions sum to

$$\mathrm{AUC}^*_{\mathrm{static}} = \frac{1}{2} + \frac{\sum_{e,f} |\pi_e - \pi_f|}{4PQ}. \tag{S2}$$

This is a finite-edge specialisation of classical bipartite ranking: the Bayes score orders the posterior positive probability, and the optimal risk depends on its Gini mean difference [Clémençon et al., 2008, Example 1]. Here that posterior is π_e when time is omitted. Our contribution is its use as an occupancy diagnostic for dynamic graph recovery, together with the temporal controls, rather than a new general optimal-ranking theorem. This is a ceiling for *constant* scores, not for dynamic predictors, and it is not a pass line — a model can localise changing edges correctly yet score below it through cross-edge ranking error.

A finite counterexample shows how far this goes: with 100 candidate edges over 100 windows — 10 always active, 10 active in half the windows, 80 never active — a constant prediction that emits scores $1, 0.5, 0$ by frequency reaches pooled AUROC 0.990, unchanged under independent time shuffling, while the per-edge temporal AUROC restricted to the changing edges is exactly 0.5: the predictor separates frequent from rare edges almost perfectly and has no temporal state-discrimination ability at all. This is not an experimental estimate; it is an exact finite construction showing that including the time axis in the scoring table does not by itself make the metric measure time information. We therefore report the pooled metric *and* the within-edge temporal AUROC $\mathrm{T{-}AUC}_\Delta$ (mean over edges in $\mathcal{E}_\Delta$ of the per-edge AUROC of $(\widehat{S}_{w,e})_w$ against $(A_{w,e})_w$, restricted to edges whose active label flips in the evaluation windows), where a constant prediction scores 0.5; the pooled metric alone can be dominated by long-run occupancy, so the two must be read together. The analogous issue — simple memory baselines and easy negatives changing rankings — is documented for dynamic link prediction [Poursafaei et al., 2022], but that task predicts links on an observed graph rather than recovering latent lagged structure from raw series, so the evaluation decomposition here is the lagged-recovery analogue rather than a direct transfer.

The earlier twenty-seed exploratory comparisons in Section S6.2 report both targets and the static ceiling in Table S17.

S3.2 State-assignment discrimination and graph contrast

The changing-edge metric has a two-factor reading that locates the remaining gap. For a two-state hard assignment $\widehat{z}_w \in \{0, 1\}$ with one fixed score map $G_{\widehat{z},e}$ per state, and true states z_w, define the state-assignment discrimination $J = \Pr(\widehat{z} = 1 \mid z = 1) - \Pr(\widehat{z} = 1 \mid z = 0)$ (the true- minus-false-positive rate under a fixed label matching) and, for a genuinely changing edge with $d_e = A_{1,e} - A_{0,e} \in \{\pm 1\}$ and predicted gap $\delta_e = G_{1,e} - G_{0,e}$, the state-contrast concordance $C_{\mathrm{edge}} = \frac{1}{|\mathcal{E}_\Delta|} \sum_{e \in \mathcal{E}_\Delta} \mathrm{sign}(d_e \delta_e)$. As long as the prediction is $\widehat{S}_{w,e} = G_{\widehat{z}_w,e}$ and the evaluation windows contain both states, each changing edge has only two predicted values, so its per-edge AUROC is fixed by the true- and false-positive rates of the assignment and by whether the two values point the right way; averaging over $\mathcal{E}_\Delta$ gives the exact identity

$$\mathrm{T{-}AUC}_\Delta = \frac{1}{2} + \frac{1}{2} J\, C_{\mathrm{edge}}. \tag{S3}$$

Swapping the predicted state numbering flips both J and C_{edge}, so the product is invariant. On the twenty core sequences the mean per-seed product $\mathbb{E}[JC_{\mathrm{edge}}] = 0.298$ reproduces mean within-edge

temporal AUROC 0.649 to 10^{-16}. This is an implementation-consistency check of an identity, not independent evidence predicting where aggregation helps. It differs from $\mathbb{E}[J]\mathbb{E}[C_{\text{edge}}] = 0.221$; the marginal means in Table S20 must not be multiplied. The two factors are *jointly estimated*, not independent levers: relabelling changes both J and the grouping that defines the state maps, hence C_{edge} too, so the oracle–K-means gap (0.814 vs 0.649) is not uniquely attributable to one factor. A conditional diagnostic along a fixed order is still informative. Holding the fitted state maps fixed and replacing the assignment by the true one gives $\frac{1}{2} + \frac{1}{2}\mathbb{E}[C_{\text{edge}}] = \frac{1}{2} + \frac{1}{2}(0.414) = 0.707$; rebuilding the maps from the true labels reaches the oracle 0.814. Along this order the gap splits as $0.814 - 0.649 = (0.707 - 0.649) + (0.814 - 0.707) \approx 0.058 + 0.107$: correcting the window assignment at fixed maps recovers about 0.058, and rebuilding the maps under the true assignment recovers about 0.107. A different intervention order would apportion the two terms differently, so these are conditional diagnostics, not unique causal contributions; in particular, correcting the labels alone is not guaranteed to recover the full oracle gain.

The pooled metric admits a parallel split. Let n_e^+, n_e^- be the positive and negative window counts of edge e. Every "window × edge" pair is either a same-edge comparison (two windows of one edge) or a cross-edge comparison (two edges), so

$$\text{AUC}_{\text{pooled}} = \omega\,\text{AUC}_{\text{same}} + (1-\omega)\,\text{AUC}_{\text{cross}}, \qquad \omega = \frac{\sum_e n_e^+ n_e^-}{\left(\sum_e n_e^+\right)\left(\sum_e n_e^-\right)}, \tag{S4}$$

where AUC_{same} is the per-edge time AUROC weighted by $n_e^+ n_e^-$ (equal to the unweighted $\text{T}-\text{AUC}_\Delta$ here because the changing edges share one state schedule) and $\text{AUC}_{\text{cross}}$ is the cross-edge ranking. On this evaluation panel $\omega = 0.0017$: only about 0.17% of the positive–negative "window × edge" pairs directly compare the same edge across two times ($\text{AUC}_{\text{same}} = 0.649$), the rest being cross-edge comparisons ($\text{AUC}_{\text{cross}} = 0.776$). This is a statement about the *composition* of the pairs, not about how much time information the metric weighs: cross-edge pairs can still depend on time. A prediction that scores every edge high only at its active times can reach a pooled AUROC of 1, and swapping the two times' scores drops it to 0, even though ω is unchanged — so ω is not an upper bound on the metric's time sensitivity. What the split does show is that the pooled metric mixes two comparison types with very different weights, which is why it cannot *by itself* diagnose within-edge temporal discrimination and must be reported with $\text{T}-\text{AUC}_\Delta$. The weight ω is fixed by the evaluation truth and sample counts, not by any method.

S3.3 Gate geometry at finite window counts

The gate's main path accepts a two-state split when the relative inertia improvement $\Delta_2 = 1 - \text{SSE}_2/\text{SSE}_1$ exceeds 0.12. With M centred window features $x_1, \ldots, x_M$ and total scatter $\text{TSS} = \sum_i \|x_i\|^2 > 0$, placing point i alone in one cluster and the remaining $M-1$ points in the other gives between-cluster scatter $\text{BSS}_i = \frac{M}{M-1}\|x_i\|^2$. Since $\max_i \|x_i\|^2 \geq \text{TSS}/M$, the globally optimal bipartition satisfies

$$\Delta_2^* \;\geq\; \frac{1}{M-1}. \tag{S5}$$

This needs no Gaussianity, no independence, no true second state, and no high-dimensional assumption. At the seven-window reference configuration this lower bound is $\frac{1}{6} \approx 0.1667 > 0.12$, so *if* the improvement is defined as the relative inertia drop and the bipartition reaches the global optimum, the main path cannot reject any non-degenerate seven-window data, whether or not a second state exists. A fixed threshold therefore loses its test meaning at small window counts: the same 0.12 that is automatic at seven windows becomes restrictive at nineteen, whose lower bound is $\frac{1}{18} \approx 0.0556$.

The bound constrains the *optimal* bipartition, not the partition K-means actually returns, so we examine that gap on the earlier exploratory core. Rebuilding the seven-window clustering features for each of the twenty switching-core sequences and enumerating all 63 inequivalent non-empty bipartitions gives an optimal relative improvement of 0.221 on average (min 0.206); the K-means-returned solution is essentially optimal (mean 0.219, within 0.014 of the optimum in every seed) and exceeds 0.12 in all 20/20 sequences. On these twenty sequences the geometry therefore carries the acceptance directly: the optimum is at least $\frac{1}{6}$, K-means is within 0.014 of it, and the threshold is below both. Two statements are deliberately kept apart: the *mathematical* claim is that a bipartition attaining the global optimum is accepted on any non-degenerate seven-window data whether or not a second state exists; the *empirical* claim is that the returned K-means solution is within 0.014 of that optimum on our twenty sequences, so the geometric problem does occur in the current implementation. The open question is not whether K-means finds a good partition but whether relative inertia improvement is the right test at all.

An independent simulation confirms the geometry. For 200 replicates of state-free Gaussian features in 168 dimensions, standardized across windows, K-means with ten restarts exceeded 0.12 in 200/200 cases at $M = 5, 7, 9$ and in 0/200 at $M = 19$, with mean improvements 0.284, 0.199, 0.154 and 0.074 respectively (the first three also verified by exhaustive bipartition enumeration). More generally, for fixed non-empty group sizes, a random assignment of windows to K groups has expected relative improvement $(K-1)/(M-1)$, so moving from one to two clusters produces a sizeable fit gain even under no state structure. This does not estimate the gate's false-acceptance rate; it isolates the geometric dependence of the threshold on the number of windows, and it predicts the earlier exploratory grid's observation that the gate accepts at seven windows. At nineteen windows (a T=3000 switching series, same D=300) the measured improvement is 0.090: below the 0.12 threshold, so the gate rejects 18/20 *switching* fits. The geometry no longer forces acceptance there (the bound drops to $\frac{1}{18} \approx 0.056$), and the switching signal at this resolution does not clear the fixed threshold on its own — a rejection is therefore not evidence that a state is absent, exactly as an acceptance is not evidence that one is present. A calibrated gate must compare the improvement to a no-state reference distribution at the same M, in the spirit of the gap statistic [Tibshirani et al., 2001], rather than to a fixed 0.12; the high-dimension/low-sample-size geometry here is well studied [Hall et al., 2005, Loog et al., 2023], and the point is to carry its consequence into the windowed state-acceptance mechanism, not to rediscover it.

S3.4 Moment-weighted window mixtures

A local regression does not estimate the mechanism active at the window centre when a window mixes states. Write the state duration as D and the full cycle as $P = 2D$ (state A for D steps, then B for D steps, repeating). For $y_t = x_t^\top \beta_{z_t} + \varepsilon_t$ with state fractions $q_{w,k}$ and conditional design second moments $Q_{w,k}$ inside a window, assuming the per-window errors are orthogonal to the design and the mixture Gram matrix $\sum_k q_{w,k} Q_{w,k}$ is invertible, the no-regularisation population least-squares coefficient is the moment-weighted mixture

$$\beta_w^* = \Big(\textstyle\sum_k q_{w,k} Q_{w,k}\Big)^{-1} \Big(\textstyle\sum_k q_{w,k} Q_{w,k} \beta_k\Big), \tag{S6}$$

which reduces to $\beta_w^* = \sum_k q_{w,k} \beta_k$ when the conditional design covariances coincide. When $h = 2D = P$ every window holds both states equally ($q_{w,1} \equiv 0.5$), so $\beta_w^* = (\beta_1 + \beta_2)/2$ for *all* windows: the population window representation is already identical across windows, and no amount of additional estimation accuracy recovers the window-centre state. This is a different failure from "too few samples" — more samples improve the estimate of a representation that has already merged the

two mechanisms. Note that "identical population coefficients" here is a statement about the population regression, not a claim that the finite-sample score distribution carries no state information at all; finite samples can still differ through estimation error.

The switched-regression and conditional denoising probes of this representation are reported in Section S6.5.

S3.5 Auxiliary stylized mean-squared-error analysis

This section is a stylized analysis of the aggregation step, not a risk bound for the full estimator. Consider two equal-size states with coefficient gap Δ, M windows, and per-window estimate variance σ^2. Assume the per-window errors are independent and conditionally mean-zero, that the misassignment mechanism is independent of the estimation errors (a fixed contamination rate alone does not exclude selection on the noise), that a fixed fraction η of windows in each group is misassigned to the other group (so the within-group contamination rate is fixed at η, not merely the group sizes), and that misassigned windows contribute the opposite state's mean. The grouped estimator averages $M/2$ windows per state:

$$\mathrm{MSE}_{\mathrm{group}} = \eta^2\Delta^2 + \frac{2\sigma^2}{M}, \tag{S7}$$

$$\mathrm{MSE}_{\mathrm{pool}} = \frac{\Delta^2}{4} + \frac{\sigma^2}{M}, \tag{S8}$$

so grouping is better exactly when

$$\left(\tfrac{1}{4} - \eta^2\right)\Delta^2 > \frac{\sigma^2}{M}. \tag{S9}$$

The tolerable label error falls as $\sqrt{1/4 - \sigma^2/(M\Delta^2)}$ and reaches zero at $\Delta^2/4 = \sigma^2/M$. A Monte Carlo check with fixed group sizes reproduces both formulas to within 0.0015 (grouped) and 0.0003 (pooled) mean squared error.

Max-selection bias. For a null edge with two independent state scores $Z_1, Z_2 \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$, $\mathbb{E}[\max(Z_1, Z_2)] = \sigma/\sqrt{\pi} \approx 0.56\sigma$ (standard normal is the σ=1 case; this is the two-state formula, not a general K-state one), while the mean has expectation zero. In a toy with 2,000 edges, 5% true, and signal 1.5σ, max aggregation lowers AUC from 0.933 to 0.901 when the edge is shared by both states, but raises it from 0.774 to 0.803 when the edge is state-specific.

Probing η with oracle labels. The theory's η is the error rate against the true state, so corrupting K-means labels does not test it directly. We therefore also corrupt the *true* window labels at rate η and aggregate with them: union AUROC falls from 0.814 to 0.794 and regime AUPRC from 0.639 to 0.556 as η goes from 0 to 0.5. The K-means curve is flatter because it starts lower; it measures estimator robustness, not the threshold.

What is not covered. Equation (S9) assumes Gaussian, equal-variance, fixed-support estimates. It is not a Lasso risk bound, does not model overlapping windows, and says nothing about AUROC. The absolute-value score discards sign, so two states with $+c$ and $-c$ on the same edge share the same absolute-coefficient representation.

S3.6 Information value, probabilistic risk, and estimation

This subsection is a general risk account, not a ranking-level explanation of the AUROC differences above: it states why available information need not improve *probabilistic prediction risk*, and connecting it to AUROC needs a separate ranking argument (given below). That the oracle label helps (0.806 versus 0.776) shows the state information has value; that the fitted clustering does not realise it shows value is not gain. For a binary edge indicator Y with nested information sets $\mathcal{I}_0 \subseteq \mathcal{I}_1$ (the state adds information), let $p_0 = \mathbb{E}[Y \mid \mathcal{I}_0]$, $p_1 = \mathbb{E}[Y \mid \mathcal{I}_1]$, and let a fitted predictor q be $\mathcal{I}_1$-measurable. Under squared (Brier) loss, orthogonality of conditional expectations gives

$$R(p_0) - R(q) = \underbrace{\mathbb{E}[(p_1 - p_0)^2]}_{\text{value of the new information}} - \underbrace{\mathbb{E}[(q - p_1)^2]}_{\text{error in how it is used}}, \tag{S10}$$

so the value term is non-negative while the whole difference can be negative. In a two-state example with a symmetric state-error rate 0.3 and an edge present in one state only, always predicting 0.5 has Brier risk 0.25, hard-coding the noisy state into 0/1 has 0.30, and the calibrated 0.3/0.7 has 0.21: the information is real, but a naive use of it is worse than ignoring it (200,000 draws give 0.2500, 0.2994, 0.2097). This is a mechanism illustration, not an identity for the absolute Lasso scores (which are not edge probabilities).

The same example shows what the decomposition does *not* explain. Under the identical setting, the AUROC of "always predict 0.5" is 0.50, while both the hard 0/1 and the calibrated 0.3/0.7 predictions reach 0.70: the soft score is the strictly increasing affine map $q_{\text{soft}} = 0.3 + 0.4\, q_{\text{hard}}$, and a strictly increasing transformation preserves the ranking and hence the AUROC. The toy therefore shows that misusing state information hurts *probabilistic* risk, not that it lowers AUROC below a static prediction — in the toy the hard use of the state already beats the constant predictor on AUROC. The decomposition is a proper-scoring-rule statement (Brier is strictly proper), not a ranking statement, so it should not be read as an explanation of the observed AUROC differences without that ranking analysis.

Two assumptions are explicit. First, the information sets must be nested, $\mathcal{I}_0 \subseteq \mathcal{I}_1$, and q must be $\mathcal{I}_1$-measurable. Second, if $\mathcal{I}_0$ already contains the full raw series and the fitted state is a deterministic function of that series, then appending the fitted state does not *enlarge* the information set; it can only change how a restricted estimator uses information that is already present. Oracle states and data-computed states must therefore be kept apart, which is why the oracle row is a *reference* (an upper target from perfect state labels, not a proven ceiling on every scoring loss — Eq. (S2) is the only ceiling stated here, and it is restricted to constant scores) and the fitted row is the object of study.

Two further estimation-side effects are worth flagging. First, clustering and then averaging the *same* window scores introduces selection: with one true state $X \sim \mathcal{N}(\mu, \sigma^2)$, splitting at μ and averaging each side yields $\mathbb{E}[X \mid X > \mu] = \mu + \sigma\sqrt{2/\pi}$ and $\mu - \sigma\sqrt{2/\pi}$ (measured ± 0.798 for $\sigma = 1$), while an independent noise replica partitioned by the same cut returns to μ (0.002/0.001); this is the selective-inference problem of clustering on the same data used for the test [Chen and Witten, 2023], though its i.i.d. Gaussian theory does not transfer verbatim to overlapping time windows. Second, refitting each contiguous run and then averaging is not the same as pooling the runs: for unregularised OLS with per-run Gram matrices Q_r, the pooled estimate is $\hat{\beta}_{\text{pooled}} = (\sum_r Q_r)^{-1} \sum_r Q_r \hat{\beta}_r$, which generally differs from the arithmetic average of the $\hat{\beta}_r$ (verified to machine precision), so run-wise averaging uses neither the sample nor the design information optimally. These point to concrete interventions — an independent noise replica for state identification, and legal row-pooled refitting — as diagnostics rather than as deployable methods.

S4 Study designs and detailed results for Studies A–C

S4.1 Paired-duration Study A and schedule Study B

Study A uses 100 new independent simulation repeats (seeds 50000–50099) at $D = 300$ and 600, with $T = 1200$ and seven windows. Study B uses 50 different repeats (seeds 60000–60049), $T = 3600$ and 23 windows, crossing both duration scales with fixed phase, random phase and irregular runs. Both retain $N = 8$, $L = 3$, window length 300, stride 150 and the fixed Lasso penalty 0.005. Their protocols and source hashes were frozen locally before execution; this is not a claim of public preregistration. Within a repeat, all conditions share the coefficient tensors, graph supports, initial history and complete innovation array. Study A additionally shares these inputs across Lasso and OLS. Independent repeats redraw the mechanism and noise; overlapping windows, edges and enumerated random partitions are not replicates.

Schedules and random streams. The fixed full-time schedule is $z(t) = \lfloor t/D \rfloor \bmod 2$, followed by the common 100-step burn-in removal. Random phase draws $q \sim \text{DiscreteUniform}\{0, \ldots, 2D - 1\}$ and uses $\lfloor (t + q)/D \rfloor \bmod 2$ before the same removal. Irregular runs start in state 0 at full-time zero, alternate states and independently draw each integer duration from $[\lceil 0.75D \rceil, \lfloor 1.25D \rfloor]$, inclusive. The retained records identify the sampled segments, the observed first/last truncations, actual switches, window-centre labels and purity of each window. Study A uses independently addressed PCG64 streams with `SeedSequence([repeat,domain])`: domains 0, 1 and 2 generate the mechanism, initial history and innovations. Study B first derives a stable 128-bit seed from its study namespace, repeat and stream domain by SHA-256, then uses the same three base-input domains. Each duration–schedule cell has an independently addressed schedule stream. Clustering uses one fixed random state within each repeat. The frozen machine-readable protocols supply the exact stream addresses; the clustering seed uses the first 32 SHA-256 bits in Study B. All fits limit BLAS/OpenMP pools to one thread.

Primary, auxiliary and secondary comparisons. For a specified hard policy, write $\Delta_D = T_{\text{hard}}(D) - T_{\text{raw}}(D)$ and $\Gamma = \Delta_{600} - \Delta_{300}$. Study A's primary family contains only four Lasso/gated-hard contrasts: both Δ_D, Γ, and hard minus size-matched random grouping at $D = 600$. They use two-sided paired repeat tests with Holm4 correction. Table S3 also gives the prespecified forced and OLS auxiliary contrasts; their raw p values do not acquire a primary-family interpretation. Study B is secondary robustness evidence: its six forced-minus-raw tests use Holm6, while the three directly paired duration differences form a separate Holm3 family. These families are not pooled, and no family correction is assigned to the descriptive gate subgroups. All reported intervals resample independent paired repeats with 10,000 percentile-bootstrap draws and seed 20261024; they are ordinary pointwise 95% intervals, not simultaneous intervals. No repeats were replaced or added after inspecting significance. At $D = 300$, the unrounded Lasso means are 0.6385365723 for gated hard and 0.7074056527 for raw, yielding paired mean -0.0688690804. Separate rounding of the two means to three decimals changes their displayed subtraction; all contrasts are calculated before that rounding.

Gate-conditioned effects and factors. Table S4 separates all-repeat gated hard output from the forced candidate. Its accepted-only mean uses each accepted repeat's own raw baseline and is conditional on this fitted selection event. At $D = 300$, only two, zero and three repeats are accepted, so those means are not used for confirmatory inference; an empty subgroup remains missing. The factors in panel (b) instead use all fifty evaluable candidates. They average the per-repeat product

Table S3: All sixteen Study A contrasts, $n = 100$ paired repeats each. Δ_D is the displayed hard policy minus raw at duration D; Γ is its paired duration difference. Rows marked P alone comprise the frozen primary Holm4 family. Dashes in the Holm column mean no adjusted p value was assigned to an auxiliary contrast, not a missing raw test. All OLS rows are auxiliary. Gated and forced coincide because every Study A gate accepts. Matched-random AUROCs are averaged within repeat after evaluating each partition separately.

Scorer	Hard policy	Contrast	n	Mean [95% interval]	p_{raw}	p_{Holm4}
Lasso	Gated^P	Δ_{300}	100	-0.069 $[-0.090, -0.048]$	5.93×10^{-9}	1.19×10^{-8}
Lasso	Gated^P	Δ_{600}	100	$+0.038$ $[0.023, 0.053]$	2.93×10^{-6}	2.93×10^{-6}
Lasso	Gated^P	Γ	100	$+0.107$ $[0.079, 0.134]$	2.99×10^{-11}	8.97×10^{-11}
Lasso	Gated^P	Hard–matched, 600	100	$+0.226$ $[0.210, 0.243]$	4.97×10^{-47}	1.99×10^{-46}
Lasso	Forced	Δ_{300}	100	-0.069 $[-0.090, -0.048]$	5.93×10^{-9}	–
Lasso	Forced	Δ_{600}	100	$+0.038$ $[0.023, 0.053]$	2.93×10^{-6}	–
Lasso	Forced	Γ	100	$+0.107$ $[0.079, 0.134]$	2.99×10^{-11}	–
Lasso	Forced	Hard–matched, 600	100	$+0.226$ $[0.210, 0.243]$	4.97×10^{-47}	–
OLS	Gated	Δ_{300}	100	-0.071 $[-0.092, -0.051]$	8.91×10^{-10}	–
OLS	Gated	Δ_{600}	100	$+0.032$ $[0.017, 0.046]$	7.93×10^{-5}	–
OLS	Gated	Γ	100	$+0.103$ $[0.076, 0.129]$	2.19×10^{-11}	–
OLS	Gated	Hard–matched, 600	100	$+0.219$ $[0.203, 0.236]$	2.28×10^{-45}	–
OLS	Forced	Δ_{300}	100	-0.071 $[-0.092, -0.051]$	8.91×10^{-10}	–
OLS	Forced	Δ_{600}	100	$+0.032$ $[0.017, 0.046]$	7.93×10^{-5}	–
OLS	Forced	Γ	100	$+0.103$ $[0.076, 0.129]$	2.19×10^{-11}	–
OLS	Forced	Hard–matched, 600	100	$+0.219$ $[0.203, 0.236]$	2.28×10^{-45}	–

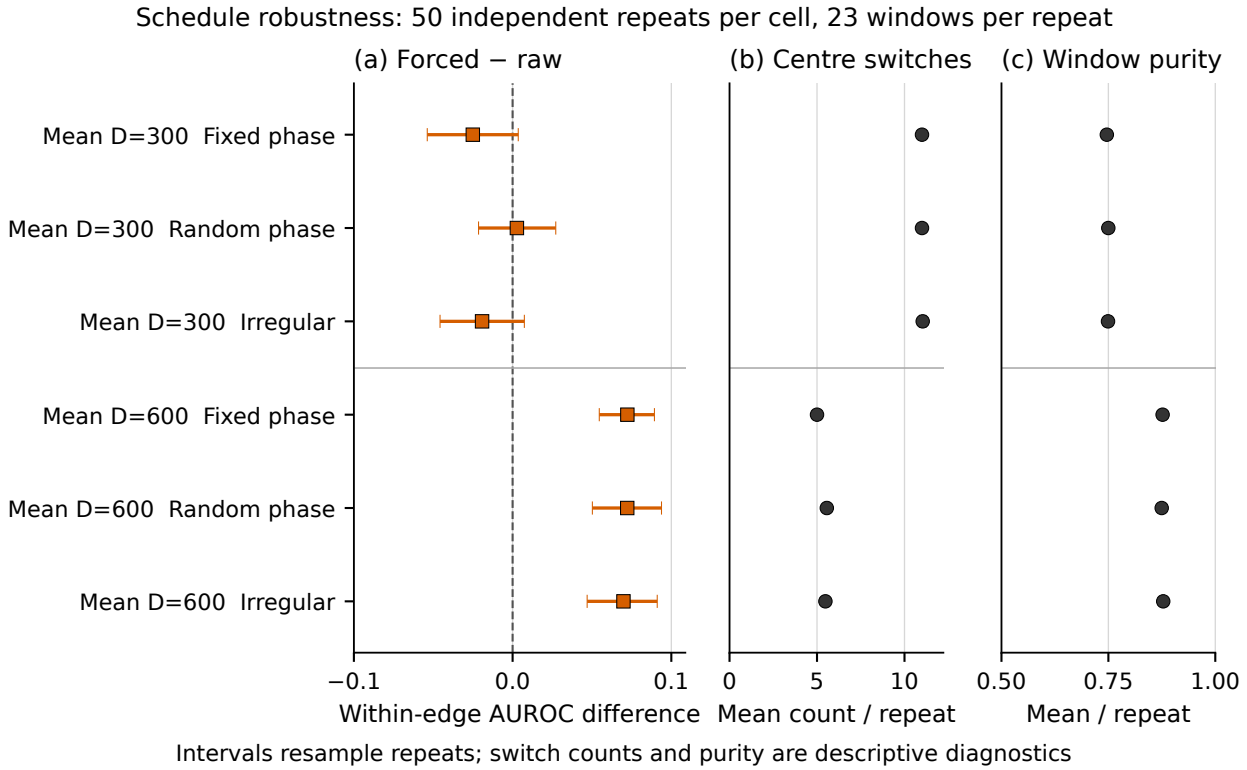


Figure S1: Study B: forced aggregation gains, realised window-centre switch counts and mean window purity for all six planned cells. Points/intervals use independent simulation repeats. The schedule comparisons are secondary robustness analyses.

JC_{edge}, preserving $\overline{T}_{\text{forced}} = \frac{1}{2} + \frac{1}{2}\overline{JC_{\text{edge}}}$, rather than multiplying two marginal means. Figure S1 retains the schedule switch-count and purity diagnostics; Table S5 gives the combined A/B factor summary.

Table S4: Study B gate coverage, conditional effects and grouping factors. Panel (a) reports accepted/observed gate decisions out of fifty planned repeats per cell; all-repeat differences retain their paired raw baselines. The final column is the descriptive forced-minus-raw mean on the n_{acc} accepted repeats, without a subgroup interval or test. A dash denotes no accepted repeat. Panel (b) reports mean factors and absolute within-edge temporal AUROC over all fifty repeats. Fixed, Random and Irregular denote the three full-time schedules defined above.

(a) Gate coverage and paired temporal gains

$\bar{D}$	Schedule	Gate	Gated–raw [95%]	Forced–raw [95%]	n_{acc}	Accepted mean
300	Fixed	2/50	−0.202 [−0.211, −0.192]	−0.025 [−0.054, 0.004]	2	−0.150
300	Random	0/50	−0.198 [−0.206, −0.189]	+0.003 [−0.021, 0.027]	0	–
300	Irregular	3/50	−0.198 [−0.207, −0.189]	−0.019 [−0.046, 0.007]	3	−0.147
600	Fixed	6/50	−0.214 [−0.245, −0.178]	+0.072 [0.055, 0.089]	6	+0.106
600	Random	6/50	−0.208 [−0.238, −0.172]	+0.072 [0.050, 0.094]	6	+0.108
600	Irregular	7/50	−0.210 [−0.241, −0.174]	+0.070 [0.047, 0.091]	7	+0.084

(b) Factors and absolute temporal AUROC on all fifty repeats

$\bar{D}$	Schedule	$\overline{J}$	$\overline{C}_{\text{edge}}$	$\overline{JC_{\text{edge}}}$	Raw	Forced
300	Fixed	0.525	0.592	0.358	0.704	0.679
300	Random	0.545	0.687	0.400	0.698	0.700
300	Irregular	0.510	0.651	0.363	0.701	0.682
600	Fixed	0.817	0.812	0.666	0.761	0.833
600	Random	0.807	0.807	0.656	0.756	0.828
600	Irregular	0.786	0.828	0.659	0.760	0.830

Table S5: Candidate-hard factor diagnostics. Every row uses all evaluable forced candidates, including gate rejections. Per-repeat products are averaged across repeats; raw reference is $2\overline{T}_{\text{raw}} - 1$. Differences among jointly estimated factors are descriptive.

Study/cell	D or $\bar{D}$	n	$\overline{J}$	$\overline{C}_{\text{edge}}$	$\overline{JC_{\text{edge}}}$	$2\overline{T}_{\text{raw}} - 1$	Forced Δ
A: Fixed phase	300	100	0.550	0.406	0.277	0.415	-0.069
A: Fixed phase	600	100	0.888	0.677	0.610	0.534	+0.038
B: Fixed phase	300	50	0.525	0.592	0.358	0.408	-0.025
B: Random phase	300	50	0.545	0.687	0.400	0.395	+0.003
B: Irregular	300	50	0.510	0.651	0.363	0.402	-0.019
B: Fixed phase	600	50	0.817	0.812	0.666	0.521	+0.072
B: Random phase	600	50	0.807	0.807	0.656	0.511	+0.072
B: Irregular	600	50	0.786	0.828	0.659	0.520	+0.070

Numerical coverage and the auxiliary scorer. All 400 Study A condition–scorer fits and 300 Study B condition fits succeed, with every within-edge temporal target evaluable. These fit counts are not counts of independent mechanisms. OLS uses the same preprocessing and lagged design

as Lasso, with `numpy.linalg.lstsq(rcond=None)` and a required full column rank; no alternative solver or regulariser is substituted. All 1,400 OLS window matrices have rank 24. Table S6 gives absolute OLS temporal scores and complete fit/window coverage. The artifact retains every window's rank and singular values for OLS, all recorded fitting/clustering warnings and the actual per-state companion radii under the unchanged rescaling rule.

Table S6: Auxiliary OLS scores and complete numerical diagnostics. Panel (a) averages 100 independent repeats at each duration; gated hard and forced hard scores coincide. Panel (b) counts condition–scorer fits and their window design matrices, not target-wise regressions or independent samples. Each Study B row combines its three schedules, with the same fifty mechanisms reused across them. Warnings combine recorded window-fitting and clustering warnings; no fit failed or OLS matrix failed its rank requirement.

(a) Auxiliary OLS absolute within-edge AUROC

D	Raw	Gated = forced hard	Matched random	Oracle	Static mean
300	0.704	0.633	0.558	0.793	0.500
600	0.765	0.797	0.577	0.847	0.500

(b) Complete fit and window-design coverage

Study	Scorer	D	Fits	Failed	Window matrices	Rank range	Warnings
A	Lasso	300	100	0	700	–	0
A	Lasso	600	100	0	700	–	0
A	OLS	300	100	0	700	24–24	0
A	OLS	600	100	0	700	24–24	0
B	Lasso	300	150	0	3450	–	0
B	Lasso	600	150	0	3450	–	0

All OLS matrices are 297×24 with rank 24. The minimum recorded singular value is 9.083; the largest recorded condition number is 3.004. Lasso ranks were not a fitting criterion and are not reported.

S4.2 Symmetric-generator and window Study C

Study C is a separately specified robustness study with 30 independent repeats (seeds 70000–70029), paired across $D = 300, 600$ and five window-size/stride settings: $(200, 100)$, $(300, 150)$, $(400, 200)$, $(300, 75)$ and $(300, 300)$. At observed length 3600 these give $M = 35, 23, 17, 45, 12$. The first three settings keep stride at half the window size; the final two vary stride with size fixed at 300. The former triplet therefore varies window size, stride and window count together. Known state count remains two; the local Lasso scorer and preprocessing are the original ones.

Both states exclude self support. For nonself positions, draw A with probability $p = 0.18$, retain its edges in B with conditional probability $o = 0.5$, and add an A-absent position with probability $p(1 - o)/(1 - p) = 0.109756\ldots$. Then the two support marginals equal p and the joint pair is exchangeable under swapping state names. Realised counts and overlap remain random. Empty graphs are allowed without resampling. Shared support retains identical signed coefficients; exclusive edges use the same independent sign and magnitude distribution in either state. These changes, together with the shared rescaling rule below, form one combined robustness control, not a factorial identification of individual contributions.

Let $B_{k,\ell}$ denote the lag-ℓ coefficient matrix in state k, distinct from the binary support tensor A. Let ρ_0, ρ_1 be initial companion radii and $a = \min\{1, 0.90/\max(\rho_0, \rho_1)\}$. A shared lagwise rescaling $B_{k,\ell} \leftarrow a^\ell B_{k,\ell}$ preserves equality of shared coefficients. The companion transformation is similar to

multiplying the initial companion matrix by a, so every state radius scales by a. When scaling is required, a recorded relative 10^{-12} safety margin is applied to a; initial/final tensors and radii are retained and numerically checked. Statewise stability does not imply arbitrary-switching stability. Unlike the earlier saved-output audit, this study retains full burn-in and observed trajectories for finite-value and growth diagnostics.

All 300 planned fits are retained, including failures and undefined targets; there is no result-dependent replacement or extension. Source/configuration hashes and internal clock records bracket protocol freeze and execution. It is a separately specified, internally frozen robustness study, not an external preregistration or a new primary confirmation of the original generator's duration contrast.

Table S7: All ten Study C window cells. h/s denotes window size/stride. Forced and raw-fallback gains are within-edge temporal AUROC minus raw, with pointwise 95% paired-repeat bootstrap intervals. The 300/150 and 300/300 long-duration cells show that removing constant fallback does not guarantee beneficial gate selection.

D	h/s	M	Accepted	Forced Δ	Raw fallback Δ
300	200/100	35	0/30	$+0.011\ [-0.018, 0.038]$	$+0.000\ [0.000, 0.000]$
300	300/150	23	3/30	$-0.047\ [-0.084, -0.006]$	$-0.018\ [-0.039, 0.000]$
300	400/200	17	8/30	$-0.139\ [-0.162, -0.112]$	$-0.045\ [-0.075, -0.019]$
300	300/75	45	12/30	$-0.028\ [-0.068, 0.011]$	$-0.049\ [-0.079, -0.021]$
300	300/300	12	11/30	$-0.089\ [-0.112, -0.061]$	$-0.022\ [-0.044, 0.000]$
600	200/100	35	22/30	$+0.072\ [0.045, 0.098]$	$+0.069\ [0.044, 0.094]$
600	300/150	23	3/30	$+0.037\ [0.002, 0.067]$	$-0.007\ [-0.027, 0.009]$
600	400/200	17	15/30	$+0.045\ [0.009, 0.077]$	$+0.034\ [0.009, 0.058]$
600	300/75	45	30/30	$+0.100\ [0.074, 0.121]$	$+0.100\ [0.074, 0.121]$
600	300/300	12	28/30	$-0.014\ [-0.049, 0.022]$	$-0.007\ [-0.039, 0.027]$

S5 Score-emission and boundary-evaluation sensitivity

These additional controls on existing outputs leave Studies A/B, their fitted outputs and their original test families unchanged. The added controls are evaluated on the retained scores with fixed $\lambda \in \{0.25, 0.5, 0.75, 1\}$; the residual family also retains the existing hard endpoint $\lambda = 0$. Every value is reported without choosing a value using the test truth. Pointwise bootstrap intervals use independent simulation repeats. These descriptive existing-output analyses do not inherit the original primary Holm labels.

S5.1 Rank-preserving controls and fallback decomposition

For any fixed edge e, a transformation $S'_{w,e} = a_e S_{w,e} + b_e$ with $a_e > 0$ preserves the signs of all positive–negative score differences and all ties. The finite pair-comparison definition of AUROC therefore gives $\mathrm{AUROC}(A_{\cdot,e}, S'_{\cdot,e}) = \mathrm{AUROC}(A_{\cdot,e}, S_{\cdot,e})$. Averaging over the same changing edges proves invariance of $\mathrm{T-AUC}_\Delta$. This elementary rank property supplies a control-design criterion rather than a new ranking theorem. For global shrinkage, $a_e = \lambda$ and $b_e = (1-\lambda)\overline{S}_e$. A cross-edge comparison instead uses

$$S^{\mathrm{shrink}}_{u,e} - S^{\mathrm{shrink}}_{v,f} = \lambda(S_{u,e} - S_{v,f}) + (1-\lambda)(\overline{S}_e - \overline{S}_f),$$

whose sign can change for $e \neq f$. Hence pooled ranking can improve or worsen without any change in within-edge temporal discrimination. At $\lambda = 0$ the temporal scores become constant, so the positive-slope invariance no longer applies. Finite-precision order/tie changes are checked explicitly rather than replaced by theoretical ranks.

For a hard candidate H, residual emission $H + \lambda(S - H)$ preserves differences between raw windows assigned to the same candidate group up to the positive factor λ. Across groups, the two hard-map offsets can alter rankings. There is no claimed monotonicity of its AUROC as λ varies. This control tests the cost of discarding residual information using the same candidate and local scores.

For each evaluable repeat let I denote acceptance, R the raw within-edge temporal AUROC and H the forced hard within-edge temporal AUROC. Constant fallback has score 1/2. The gated hard gain is $I(H - R) + (1 - I)(1/2 - R)$; raw fallback has gain $I(H - R)$. Their difference is $(1 - I)(R - 1/2)$. On a common evaluable repeat set,

$$\overline{\Delta}_{\text{raw fallback}} = \widehat{\Pr}(I = 1)\,\overline{(H - R) \mid I = 1},$$

with a zero overall gain when no candidate is accepted. Conditional effects use their own raw baseline; acceptance among all fitted repeats is reported separately if metric coverage differs. The two fallback policies use the same fitted candidate and gate decision.

Table S8: Auxiliary OLS score controls on Study A's retained fits. Every fixed non-endpoint weight is shown, alongside raw and forced hard. Scores are within-edge temporal AUROCs; $\lambda = 1$ duplicates raw. These additional existing-output comparisons have no primary-family interpretation.

Emission	Pooled AUROC, $D = 300$	Within-edge, 300	Pooled AUROC, 600	Within-edge, 600
Raw	0.724	0.704	0.770	0.765
Forced hard	0.778	0.633	0.837	0.797
Shrink 0.25	0.801	0.704	0.841	0.765
Shrink 0.5	0.778	0.704	0.822	0.765
Shrink 0.75	0.750	0.704	0.796	0.765
Residual 0.25	0.776	0.712	0.832	0.821
Residual 0.5	0.763	0.712	0.817	0.805
Residual 0.75	0.745	0.709	0.794	0.785

Table S9: Study B fallback separation with the acceptance decision unchanged. All differences use the same repeats and their own raw baseline; intervals are pointwise paired-repeat 95% percentile-bootstrap intervals. Repeated gated/forced hard contrasts use the original Study B summaries (seed 20261024); raw fallback retains the additional-controls stream (seed 20261025). A zero raw-fallback gain with no acceptance is an exact emission identity.

D	Schedule	Accepted	Gated Δ	Raw fallback Δ	Forced Δ
300	Fixed	2/50	-0.202 $[-0.211, -0.192]$	-0.006 $[-0.016, 0.000]$	-0.025 $[-0.054, 0.004]$
300	Random	0/50	-0.198 $[-0.206, -0.189]$	$+0.000$ $[0.000, 0.000]$	$+0.003$ $[-0.021, 0.027]$
300	Irregular	3/50	-0.198 $[-0.207, -0.189]$	-0.009 $[-0.020, 0.000]$	-0.019 $[-0.046, 0.007]$
600	Fixed	6/50	-0.214 $[-0.245, -0.178]$	$+0.013$ $[0.003, 0.024]$	$+0.072$ $[0.055, 0.089]$
600	Random	6/50	-0.208 $[-0.238, -0.172]$	$+0.013$ $[0.004, 0.024]$	$+0.072$ $[0.050, 0.094]$
600	Irregular	7/50	-0.210 $[-0.241, -0.174]$	$+0.012$ $[0.003, 0.022]$	$+0.070$ $[0.047, 0.091]$

S5.2 Boundary slots and stable-edge score changes

Within-edge temporal AUROC ranks active versus inactive windows separately for each changing edge. To assess a distinct time-local target, let $c_w = \lfloor (a_w + b_w - 1)/2 \rfloor$ be window centres and define adjacent-centre slots $I_w = (c_w, c_{w+1}]$. A nonself edge's binary slot truth is one if its true support changes at least once in that slot; its predicted change strength is $|\widehat{S}_{w+1,e} - \widehat{S}_{w,e}|$. Multiple support events in a slot collapse to one target, with their counts retained. Events before the first centre or after the last centre are outside this target. This gives stride-resolution change detection, not a subwindow boundary-distance estimate or a thresholded support-recovery guarantee.

Positive-change average precision (AP) orders strictly positive change strengths, processes each tied strength as one group, and sums precision times the increment in recall. Zero strength abstains, so a constant output has AP zero when the target is evaluable. Edges with no covered boundary or no negative slot have undefined AP; their counts are reported rather than assigned zero or one. This convention differs from ordinary AP that includes a final zero-score group. Absolute AP depends on boundary prevalence and stride; comparisons with raw use the same edge set and slot grid.

For a separate amplitude-based check, one threshold per fit is the empirical 95th percentile (linear quantile interpolation) of all raw nonself adjacent-window change strengths. Every method uses that same threshold and predicts an event only when its change strength strictly exceeds it. Boundary-slot F1 is calculated from TP, FP and FN, with zero assigned when there is a true boundary but no prediction. Its edge macro-average includes covered changing edges even if every slot is positive, with that coverage flagged. This raw-derived threshold uses no graph labels and is a fixed descriptive reference, not a calibrated false-alarm probability.

For edges that stay present or absent throughout the observed truth, we also report threshold exceedances divided by edge–slot opportunities and the fraction of stable edges with any exceedance. Persistent and absent edges are reported separately. Constant output has no stable-edge variation but also detects no genuine boundary; both targets must be inspected. The full definitions, zero-threshold cases, missing coverage and paired repeat outputs are retained with the supplementary numerical files.

S5.3 Post hoc standard AP and the zero-score group

The following analyses reuse the completed A/B/C fitted outputs. Their fixed quantile grid and residual weights are descriptive choices made after those results were available; they do not extend the original confirmatory families.

For one eligible changing edge with R covered slots, let $y_r \in \{0, 1\}$ indicate a support change in slot r, and let $a_r \geq 0$ be the absolute emitted-score change. As in Section S5.2, both positive and negative slots are required for AP. Define $\pi_{\text{boundary}} = R^{-1} \sum_r y_r$ and $R_+ = \sum_r y_r \mathbf{1}\{a_r > 0\} / \sum_r y_r$. Standard non-interpolated AP includes every distinct score level, with whole tie groups [scikit-learn developers]. The original positive-change AP omits only the final zero-strength group. That group's precision is π_{boundary} and its recall increment is $1 - R_+$, giving the exact per-edge identity

$$\text{AP}_{\text{standard}} = \text{AP}_+ + \pi_{\text{boundary}}(1 - R_+). \tag{S11}$$

For constant zero changes, standard AP equals prevalence while $\text{AP}_+ = 0$. We report both conventions on the same eligible edges and average the per-edge zero-group increments, rather than multiply marginal averages. Undefined targets remain missing under both definitions.

Table S10: Boundary-slot evaluation on all Lasso conditions. AP uses positive change strength with zero-strength abstention; F1 uses the common raw-derived 95th-percentile threshold. Every difference is paired with raw on the same slot grid, with pointwise 95% bootstrap intervals. The $D = 300, h/s = 300/300$ grid has no negative boundary slot, so AP is undefined in all 30 repeats; the dash preserves that coverage rather than imputing an accuracy.

Condition $(D, h/s)$	n_{AP}	Raw AP	Forced–raw AP	Raw F1	Forced–raw F1
A 300, Fixed	100	0.724	-0.230 $[-0.296, -0.162]$	0.232	-0.165 $[-0.179, -0.151]$
A 600, Fixed	100	0.508	$+0.184$ $[0.096, 0.268]$	0.168	$+0.021$ $[-0.005, 0.047]$
B 300, Fixed	50	0.656	-0.247 $[-0.310, -0.184]$	0.243	-0.209 $[-0.220, -0.197]$
B 300, Random	50	0.632	-0.218 $[-0.274, -0.163]$	0.230	-0.185 $[-0.196, -0.173]$
B 300, Irregular	50	0.629	-0.256 $[-0.320, -0.191]$	0.220	-0.181 $[-0.195, -0.167]$
B 600, Fixed	50	0.421	$+0.068$ $[-0.003, 0.144]$	0.209	-0.077 $[-0.093, -0.060]$
B 600, Random	50	0.438	$+0.074$ $[-0.006, 0.155]$	0.209	-0.068 $[-0.088, -0.048]$
B 600, Irregular	50	0.433	-0.022 $[-0.092, 0.054]$	0.207	-0.087 $[-0.105, -0.069]$
C 300, 200/100	30	0.416	-0.238 $[-0.282, -0.190]$	0.181	-0.143 $[-0.155, -0.130]$
C 300, 300/150	30	0.664	-0.295 $[-0.384, -0.200]$	0.268	-0.225 $[-0.248, -0.200]$
C 300, 400/200	30	0.770	-0.568 $[-0.617, -0.515]$	0.240	-0.234 $[-0.245, -0.222]$
C 300, 300/75	30	0.325	-0.154 $[-0.201, -0.103]$	0.149	-0.090 $[-0.109, -0.070]$
C 300, 300/300	0	–	–	0.195	-0.187 $[-0.199, -0.176]$
C 600, 200/100	30	0.288	-0.087 $[-0.145, -0.024]$	0.158	-0.089 $[-0.104, -0.074]$
C 600, 300/150	30	0.439	-0.055 $[-0.132, 0.026]$	0.230	-0.100 $[-0.121, -0.080]$
C 600, 400/200	30	0.551	$+0.109$ $[0.008, 0.206]$	0.279	-0.083 $[-0.117, -0.050]$
C 600, 300/75	30	0.225	$+0.089$ $[-0.005, 0.190]$	0.139	$+0.027$ $[-0.009, 0.064]$
C 600, 300/300	30	0.631	-0.175 $[-0.260, -0.088]$	0.250	-0.171 $[-0.189, -0.154]$

Table S11: Stable-edge change-strength exceedance rates, split into persistent and absent support. Each fit's unlabelled raw-derived threshold is shared by every method. C shows the original 300/150 window setting; the full grid and all fixed residual/shrinkage weights are in the numerical files. These rates assess score variation and are not calibrated binary-support false alarm probabilities.

Condition	Persistent: raw	forced	raw fallback	Absent: raw	forced	raw fallback
A 300, Fixed	0.141	0.026	0.026	0.024	0.001	0.001
A 600, Fixed	0.140	0.030	0.030	0.030	0.000	0.000
B 300, Fixed	0.130	0.007	0.125	0.024	0.000	0.024
B 300, Random	0.141	0.013	0.141	0.022	0.000	0.022
B 300, Irregular	0.143	0.008	0.134	0.022	0.000	0.021
B 600, Fixed	0.133	0.022	0.119	0.028	0.000	0.026
B 600, Random	0.136	0.021	0.124	0.027	0.000	0.025
B 600, Irregular	0.131	0.020	0.115	0.028	0.000	0.025
C 300, 300/150	0.065	0.000	0.057	0.027	0.000	0.023
C 600, 300/150	0.072	0.000	0.065	0.031	0.000	0.028

S5.4 Fixed quantiles and amplitude references

For each fitted case, the quantile grid is $q \in \{0.80, 0.90, 0.95, 0.975, 0.99\}$. At each q we report two references: the quantile of all nonself raw edge-slot absolute changes, shared across methods, and each method's own quantile over its corresponding changes. Both use linear quantile interpolation and strict exceedance. Quantiles do not use support truth; no best-q or best-F1 rule is evaluated. Method-relative quantiles remove uniform positive amplitude scaling in exact arithmetic. They do not match independently calibrated false-positive rates: different tie masses and shape changes can still change exceedance rates. The complete F1 and stable-edge curves retain all fixed operating points.

A deterministic example distinguishes amplitude from order. For truth $(1, 0, 0, 0)$, changes $(1, 0.2, 0.1, 0)$ give F1 1 at threshold 0.5; multiplying every change by 0.1 gives F1 0 at the same threshold while leaving AP and ranking unchanged. This is a definition check, not simulation evidence. The method-relative reference addresses this particular ambiguity; it is not an oracle threshold and does not isolate every source of detection error.

Complete sensitivity displays. Tables S12 and S13 report all eighteen Lasso condition cells, including the zero-eligible-AP cell. The full numerical outputs also retain the two auxiliary OLS cells and every original method/weight. Intervals here use common within-cell bootstrap resamples for the sensitivity comparisons; small differences from Section S5.2's intervals reflect only its earlier Monte Carlo stream. The underlying AP_+, raw-q95 F1 and samples are unchanged.

Table S12: Hard-minus-raw AP sensitivity. Standard AP includes the zero-strength group; positive-change AP abstains there. Differences and pointwise 95% intervals use unrounded paired repeats. n_{AP} is the evaluable paired-repeat count; dashes in the zero-eligible cell remain missing, not zero.

Condition	n_{AP}	Raw AP_{std}	ΔAP_+	ΔAP_{std}
A 300, Fixed	100	0.725	−0.230 [−0.296, −0.161]	+0.006 [−0.030, 0.043]
A 600, Fixed	100	0.509	+0.184 [0.098, 0.267]	+0.229 [0.156, 0.299]
B 300, Fixed	50	0.657	−0.247 [−0.310, −0.184]	−0.025 [−0.057, 0.010]
B 300, Random	50	0.633	−0.218 [−0.272, −0.161]	+0.002 [−0.027, 0.033]
B 300, Irregular	50	0.630	−0.256 [−0.320, −0.190]	−0.013 [−0.046, 0.024]
B 600, Fixed	50	0.421	+0.068 [−0.005, 0.144]	+0.140 [0.080, 0.202]
B 600, Random	50	0.439	+0.074 [−0.006, 0.156]	+0.154 [0.089, 0.221]
B 600, Irregular	50	0.434	−0.022 [−0.092, 0.052]	+0.077 [0.022, 0.136]
C 300, 200/100	30	0.417	−0.238 [−0.282, −0.190]	−0.043 [−0.069, −0.012]
C 300, 300/150	30	0.665	−0.295 [−0.384, −0.198]	−0.037 [−0.084, 0.015]
C 300, 400/200	30	0.772	−0.568 [−0.618, −0.515]	−0.074 [−0.089, −0.059]
C 300, 300/75	30	0.325	−0.154 [−0.202, −0.103]	+0.011 [−0.022, 0.045]
C 300, 300/300	0	–	–	–
C 600, 200/100	30	0.289	−0.087 [−0.145, −0.024]	−0.006 [−0.054, 0.047]
C 600, 300/150	30	0.440	−0.055 [−0.132, 0.025]	+0.040 [−0.020, 0.104]
C 600, 400/200	30	0.552	+0.109 [0.009, 0.207]	+0.175 [0.098, 0.250]
C 600, 300/75	30	0.226	+0.089 [−0.003, 0.187]	+0.146 [0.065, 0.234]
C 600, 300/300	30	0.633	−0.175 [−0.260, −0.086]	−0.019 [−0.072, 0.040]

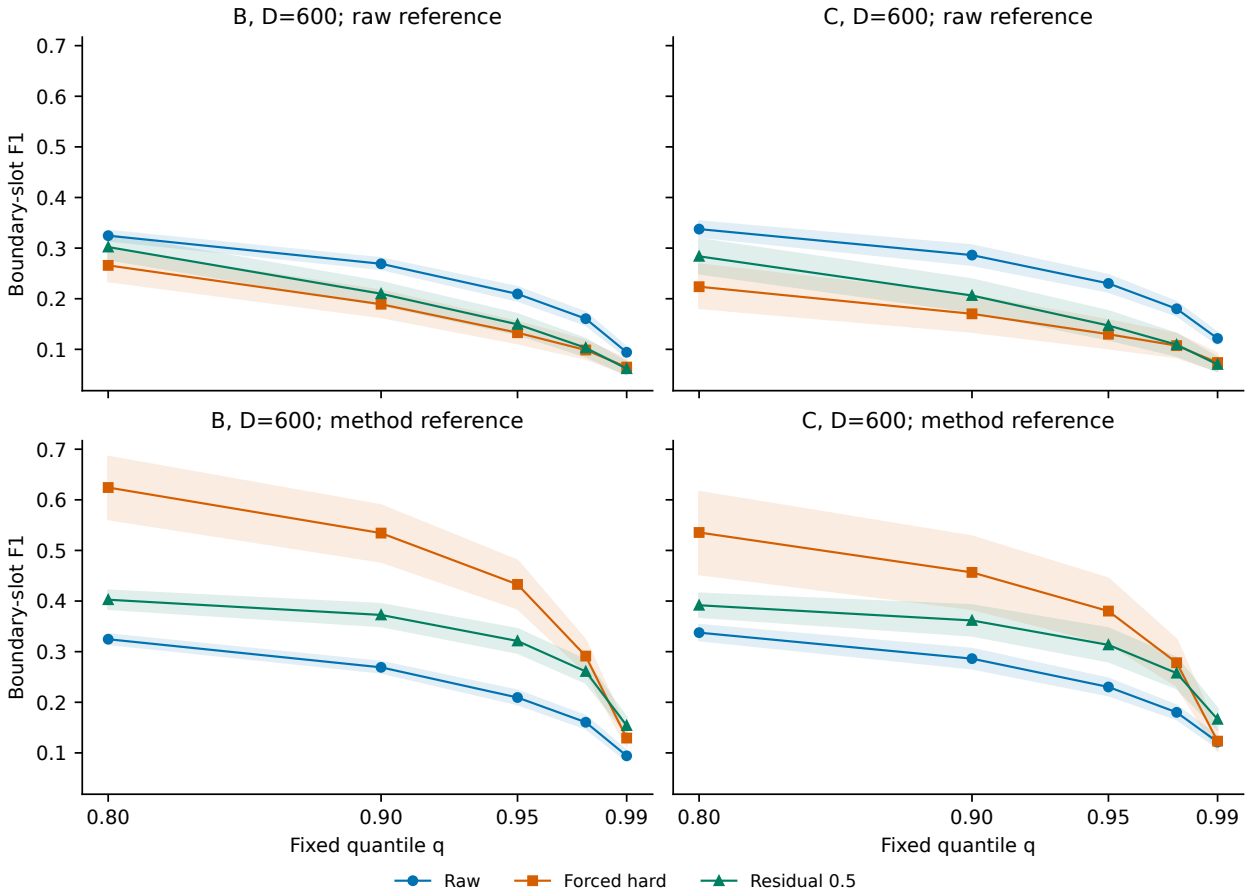


Figure S2: Fixed-quantile F1 sensitivity in the B fixed schedule and C central grid at $D = 600$. Rows change the amplitude reference; columns hold inputs and slot grids fixed. Bands are pointwise 95% repeat-bootstrap intervals, not simultaneous bands. Every tested quantile is retained, with no best-F1 selection or independent false-alarm calibration.

Table S13: Hard-minus-raw boundary-slot F1 sensitivity at fixed q95. F1 uses the shared raw or each method's own amplitude reference. Differences and pointwise 95% intervals use unrounded paired repeats: 100 per A cell, 50 per B cell and 30 per C cell. The all-positive-slot C cell retains all thirty evaluable F1 repeats even though its AP is undefined.

Condition	$\Delta\mathrm{F1}_{\mathrm{raw}}$	$\Delta\mathrm{F1}_{\mathrm{self}}$
A 300, Fixed	-0.165 $[-0.179, -0.151]$	-0.015 $[-0.040, 0.010]$
A 600, Fixed	$+0.021$ $[-0.005, 0.046]$	$+0.262$ $[0.207, 0.317]$
B 300, Fixed	-0.209 $[-0.220, -0.198]$	-0.010 $[-0.035, 0.016]$
B 300, Random	-0.185 $[-0.196, -0.173]$	$+0.024$ $[0.004, 0.043]$
B 300, Irregular	-0.181 $[-0.194, -0.167]$	$+0.016$ $[-0.009, 0.040]$
B 600, Fixed	-0.077 $[-0.093, -0.060]$	$+0.223$ $[0.178, 0.268]$
B 600, Random	-0.068 $[-0.088, -0.048]$	$+0.218$ $[0.175, 0.262]$
B 600, Irregular	-0.087 $[-0.105, -0.069]$	$+0.177$ $[0.130, 0.223]$
C 300, 200/100	-0.143 $[-0.155, -0.131]$	$+0.017$ $[-0.012, 0.046]$
C 300, 300/150	-0.225 $[-0.248, -0.200]$	-0.036 $[-0.077, 0.003]$
C 300, 400/200	-0.234 $[-0.245, -0.222]$	-0.101 $[-0.128, -0.074]$
C 300, 300/75	-0.090 $[-0.109, -0.069]$	$+0.071$ $[0.026, 0.117]$
C 300, 300/300	-0.187 $[-0.199, -0.176]$	-0.054 $[-0.069, -0.038]$
C 600, 200/100	-0.089 $[-0.104, -0.074]$	$+0.116$ $[0.066, 0.169]$
C 600, 300/150	-0.100 $[-0.122, -0.079]$	$+0.150$ $[0.093, 0.206]$
C 600, 400/200	-0.083 $[-0.118, -0.050]$	$+0.196$ $[0.143, 0.248]$
C 600, 300/75	$+0.027$ $[-0.009, 0.064]$	$+0.274$ $[0.195, 0.351]$
C 600, 300/300	-0.171 $[-0.188, -0.153]$	$+0.020$ $[-0.017, 0.059]$

S5.5 Within-group and cross-group residual pair credit

Let $\mathcal{P}_e = \{(u, v) : A_{u,e} = 1, A_{v,e} = 0\}$ and $c(d) = \mathbf{1}\{d > 0\} + \frac{1}{2}\mathbf{1}\{d = 0\}$. For a residual weight λ, write $d^{(\lambda)}_{uv,e} = (1 - \lambda)(H_{u,e} - H_{v,e}) + \lambda(S_{u,e} - S_{v,e})$. Partition $\mathcal{P}_e$ by equality of the candidate group labels, and define

$$G_g(\lambda) = \frac{1}{|\mathcal{E}_\Delta|} \sum_{e \in \mathcal{E}_\Delta} \frac{\sum_{(u,v) \in \mathcal{P}^g_e} [c(d^{(\lambda)}_{uv,e}) - c(d^{(0)}_{uv,e})]}{n^+_e n^-_e}, \quad g \in \{\text{within}, \text{cross}\}. \tag{S12}$$

Each component uses the edge's total positive–negative pair denominator, then the same macro average as within-edge temporal AUROC. Hence $G_{\text{within}} + G_{\text{cross}} = T_{\text{residual}} - T_{\text{hard}}$ exactly in the finite score representation, up to reported floating-point differences. Within a candidate group, $d^{(0)} = 0$ and every positive λ restores raw ordering; across groups both nonzero hard contrasts and tied map values are retained in the accounting. We report all original weights and a fixed 10^{-8} diagnostic weight, together with lost/new ties and reversed comparisons. A near-zero rank jump is not evidence of robust thresholded detection or improved latent-state inference.

Table S14: Residual pair-credit accounting on A and the central C grid, Lasso. Mean within/cross increments sum to residual-minus-hard within-edge temporal AUROC before rounding. Each edge uses its total positive–negative pair denominator before macro averaging. The near-zero weight is diagnostic; all fixed weights and auxiliary conditions remain in the numerical files. At $\lambda = 1$, the C total-contrast interval is the sign-reversed original C forced-minus-raw interval; within/cross component intervals retain their decomposition stream.

Study, D	λ	Forced hard	Residual	Within-group Δ	Cross-group Δ	Total Δ
A, 300	10^{-8}	0.639	0.707	$+0.068$ $[0.059, 0.078]$	$+0.000$ $[0.000, 0.000]$	$+0.068$ $[0.059, 0.078]$
A, 300	0.25	0.639	0.718	$+0.068$ $[0.059, 0.078]$	$+0.011$ $[0.006, 0.016]$	$+0.079$ $[0.066, 0.092]$
A, 300	0.5	0.639	0.716	$+0.068$ $[0.059, 0.078]$	$+0.009$ $[0.001, 0.017]$	$+0.078$ $[0.061, 0.094]$
A, 300	1	0.639	0.707	$+0.068$ $[0.059, 0.078]$	$+0.000$ $[-0.013, 0.013]$	$+0.069$ $[0.048, 0.090]$
A, 600	10^{-8}	0.805	0.830	$+0.025$ $[0.018, 0.032]$	$+0.000$ $[0.000, 0.000]$	$+0.025$ $[0.018, 0.032]$
A, 600	0.25	0.805	0.824	$+0.025$ $[0.018, 0.032]$	-0.006 $[-0.011, -0.002]$	$+0.019$ $[0.009, 0.028]$
A, 600	0.5	0.805	0.809	$+0.025$ $[0.018, 0.032]$	-0.021 $[-0.027, -0.015]$	$+0.003$ $[-0.008, 0.015]$
A, 600	1	0.805	0.767	$+0.025$ $[0.018, 0.032]$	-0.063 $[-0.072, -0.054]$	-0.038 $[-0.053, -0.023]$
C, 300	10^{-8}	0.661	0.740	$+0.079$ $[0.066, 0.092]$	$+0.000$ $[0.000, 0.000]$	$+0.079$ $[0.066, 0.092]$
C, 300	0.25	0.661	0.745	$+0.079$ $[0.066, 0.092]$	$+0.005$ $[-0.007, 0.016]$	$+0.084$ $[0.062, 0.105]$
C, 300	0.5	0.661	0.735	$+0.079$ $[0.066, 0.092]$	-0.005 $[-0.024, 0.013]$	$+0.074$ $[0.044, 0.102]$
C, 300	1	0.661	0.707	$+0.079$ $[0.066, 0.092]$	-0.032 $[-0.061, -0.005]$	$+0.047$ $[0.006, 0.084]$
C, 600	10^{-8}	0.797	0.849	$+0.052$ $[0.038, 0.067]$	$+0.000$ $[0.000, 0.000]$	$+0.052$ $[0.038, 0.067]$
C, 600	0.25	0.797	0.841	$+0.052$ $[0.038, 0.067]$	-0.008 $[-0.016, 0.001]$	$+0.044$ $[0.026, 0.063]$
C, 600	0.5	0.797	0.815	$+0.052$ $[0.038, 0.067]$	-0.034 $[-0.047, -0.021]$	$+0.017$ $[-0.005, 0.042]$
C, 600	1	0.797	0.761	$+0.052$ $[0.038, 0.067]$	-0.088 $[-0.109, -0.068]$	-0.037 $[-0.067, -0.002]$

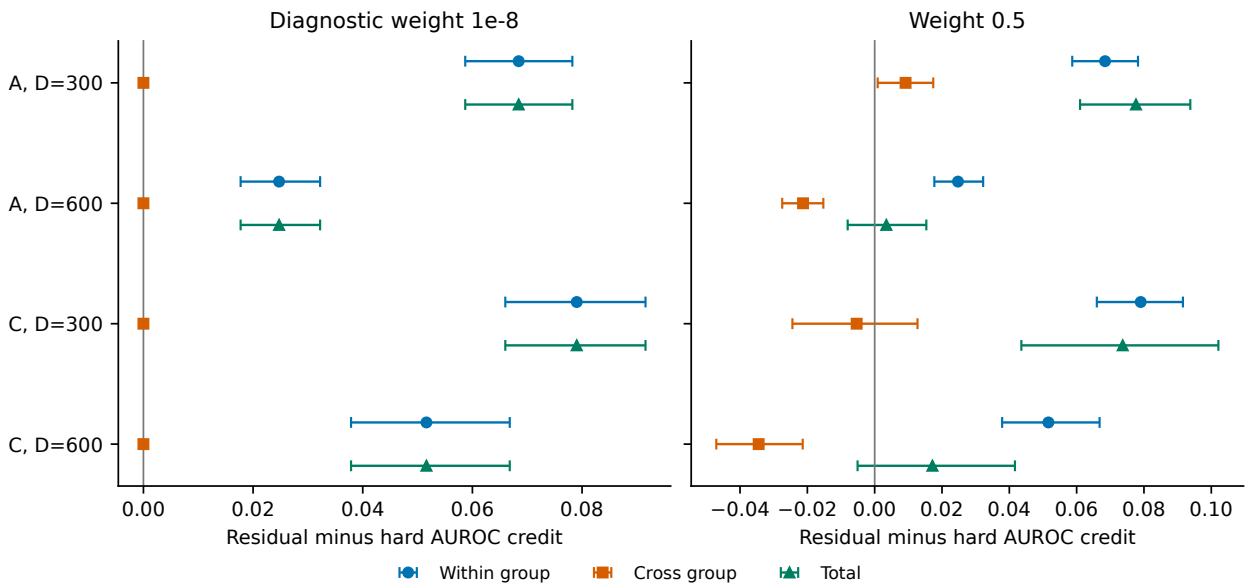


Figure S3: Residual within-edge temporal AUROC increments from within-group and cross-group comparisons. Near-zero residuals break hard-output ties without changing cross-group credit on these retained arrays. At weight 0.5, cross-group reordering can offset that gain. Pointwise intervals use repeat resampling.

S6 Additional comparisons and exploratory analyses

S6.1 Earlier exploratory mechanism grid

Design. The earlier exploratory grid has a seven-window reference configuration, one-factor scans on eight axes, two interactions, and the control family: 35 configurations with twenty seeds each, at $N = 8$, $L = 3$, $h = 300$, stride 150, $T = 1200$ unless stated. All fits supply a state count; the stationary condition is a gate-stress condition with $K_{\text{candidate}} = 2$ and $K_{\text{true}} = 1$, and is labelled as such.

Stability of the switching path. State tensors whose initial radius exceeds 0.90 are rescaled as specified in Section S2.1; this does not guarantee stability of an arbitrary switching path. Across 200 seeds the largest absolute value is 0.536, the largest ratio of that value to the series standard deviation is 4.97, the 95th percentile of the per-state variance ratio is 1.10, and no trajectory exceeds the blow-up threshold. The largest recorded per-state companion radius in this audit is 0.6505, below the rescaling trigger. The generator is therefore empirically stable on the paths it is used for, which is weaker than a joint-stability guarantee.

State duration and purity. At state duration $D = 150$ (P=300) every window straddles a boundary (purity 0.500), union AUROC is 0.756 and regime AUPRC 0.465; at $D = 900$ purity is 0.929, union 0.859 and regime 0.718. Lengthening the *switching* series to $T = 3000$ (same D=300, now 19 windows) lowers the measured relative-inertia improvement to 0.090, below the 0.12 threshold, so the gate rejects the two-state partition in 18 of 20 fits — a rejection of a *true* switch, consistent with the geometric account (Section S3.3) that a fixed threshold has no test meaning across window counts.

Support retention. Union AUROC rises from 0.768 at configured retention $o = 0$ to 0.862 at $o = 1$, and regime AUPRC from 0.530 to 0.655. Measurement noise from 0 to 0.4 moves union AUROC from 0.801 to 0.777 and regime AUPRC from 0.587 to 0.535; base-support probability p from 0.10 to 0.30 moves union AUROC from 0.786 to 0.803 and regime AUPRC from 0.499 to 0.666.

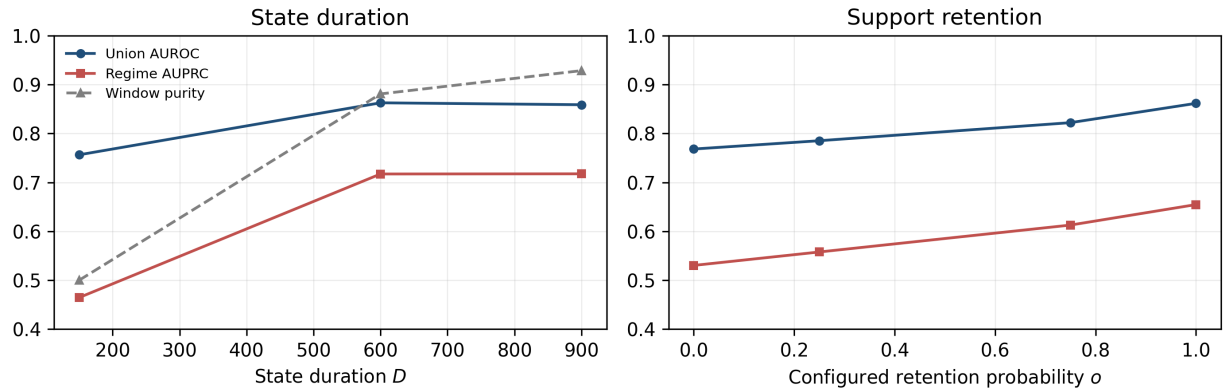


Figure S4: Recovery against state duration (left) and configured support retention (right). The left abscissa is state duration D (the full cycle is $2D$), and the right abscissa is retention probability o, not each draw's realised overlap. Window purity rises with duration and tracks both scores.

Control taxonomy. Table S15 groups the controls by what they change and reports the two structural counts. Three points matter. First, lag shift is a genuine lag-specific structure change: the pair-level graph is unchanged, so pair AUROC (0.875) exceeds lag AUROC (0.784), and state ARI (0.407) is the highest in the table. Second, the mechanism family has $K_{\text{dyn}} = 2$ but $K_{\text{graph}} = 1$: a sign flip leaves the absolute coefficients identical and drives regime AUPRC to 0.462 with ARI 0.153, and resampled magnitudes under a shared support leave ARI at -0.041. Because the two states share one binary graph, the regime AUPRC in these rows scores recovery against a single support and does not test whether the method distinguished the dynamical states; state-label alignment (ARI) alone carries that question. Third, the nuisance family also has $K_{\text{dyn}} = 2$ with $K_{\text{graph}} = 1$: mean-only is the worst case for union recovery (0.640), while variance-only (0.881) and a state-specific trend (0.672) disturb the ranking less, and none of them change the support. The gate accepts a two-state split in all eight conditions.

S6.2 Earlier aggregation comparisons across targets

Every method in this section consumes the same window-level Lasso scores; only the grouping and aggregation rule changes. Table S16 reports union, regime and pooled AUROC on the same scale.

On the switching setting, window mean reaches 0.804 union AUROC, window max 0.804, bagging 0.800, random balanced grouping 0.798, K-means 0.801, GMM responsibility-weighted maps with argmax emission 0.789, and temporal DP 0.795; the spread is 0.015. Regime AUROC stays in the same narrow band (0.786 K-means, 0.783 random, 0.805 oracle labels, 0.777 soft). The pooled AUROC is lower overall, but the static predictors – which emit the same graph at every window – reach 0.795 (window mean), 0.791 (window maximum) and 0.792 (bagging), already above fitted K-means (0.776), random grouping (0.773) and temporal DP (0.765); only oracle labels (0.806) improve on a constant graph.

Table S15: Controls grouped by what they change. Only the first block changes the lag-specific support. K_{dyn} counts dynamical mechanisms, K_{graph} counts binary supports; Pooled denotes the mixed-ranking, time-indexed AUROC.

Category	Condition	K_{dyn}	K_{graph}	Lag AUROC	Pair AUROC	Regime AUPRC	Pooled AUROC	Assignment ARI
structure change	Switching (reference)	2	2	0.801	0.792	0.587	0.776	+0.292
structure change	Lag shift	2	2	0.784	0.875	0.567	0.761	+0.407
mechanism change	Same support	2	1	0.847	0.824	0.634	0.810	-0.041
mechanism change	Sign flip	2	1	0.764	0.705	0.462	0.728	+0.153
nuisance change	Variance only	2	1	0.881	0.861	0.701	0.839	-0.096
nuisance change	Mean only	2	1	0.640	0.604	0.373	0.649	-0.143
nuisance change	Trend	2	1	0.672	0.628	0.402	0.660	-0.148
no change	Stationary	1	1	0.908	0.883	0.775	0.871	+0.000

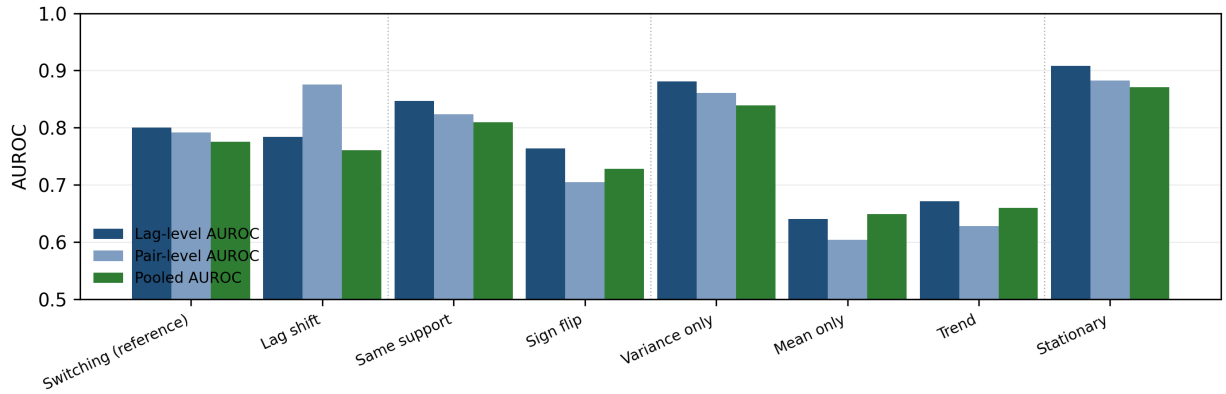


Figure S5: Lag-level, pair-level and pooled AUROC by control family. The lag-shift condition is easier at the pair level and harder at the lag level, and it has the highest state ARI: it is a real lag-specific change.

Table S17 shows this pooled reading is incomplete. The within-edge temporal AUROC $\mathrm{T}-\mathrm{AUC}_{\Delta}$ isolates within-edge temporal discrimination from long-run occupancy: K-means scores 0.649 (bootstrap 95% CI $[0.594, 0.705]$), above the 0.5 of every constant prediction and the 0.570 of random grouping, while oracle labels reach 0.814 and temporal DP 0.562. The raw window scores, used with no grouping, reach 0.716 (CI $[0.694, 0.738]$) — *above* K-means. The within-edge temporal discrimination signal therefore already lives in the local scores, and at this core setting state aggregation compresses it rather than creating it, even as it improves the pooled metric (0.776 vs 0.731). On the pooled target the fitted clustering does not beat the static controls, but on the changing-edge target it beats the constant prediction and the random grouping while trailing both the raw window scores and the oracle reference: the two targets measure different abilities, and neither alone reports both.

Table S16: Aggregation controls on the switching and stationary settings (twenty seeds). All rows share the same window scores. The GMM row weights state maps softly but uses argmax responsibility for emission. U is union, R is regime-specific, and Pooled is the mixed-ranking, time-indexed AUROC; "–" marks methods without state scores.

	Switching			Stationary		
Method	U-AUROC	R-AUROC	Pooled AUROC	U-AUROC	R-AUROC	Pooled AUROC
Window mean	0.804	–	0.795	0.919	–	0.919
Window max	0.804	–	0.791	0.893	–	0.893
Bagged mean	0.800	–	0.792	0.918	–	0.918
Random balanced	0.798	0.783	0.773	0.910	0.893	0.887
Gated hard	0.801	0.786	0.776	0.908	0.886	0.871
GMM weighted + argmax	0.789	0.777	0.776	0.906	0.896	0.876
Temporal DP	0.795	0.775	0.765	0.909	0.889	0.873
Oracle label	0.814	0.805	0.806	0.919	0.919	0.919
Oracle refit	0.844	–	–	0.954	–	–

Table S17: Pooled AUROC against the within-edge temporal AUROC $\mathrm{T}-\mathrm{AUC}_{\Delta}$ on the switching setting (twenty seeds; each sequence has 27.3 changing edges on average). "Raw window" emits each window's own scores with no grouping. Constant predictions score 0.5 on the changing-edge metric by construction; the static ceiling (Eq. S2) is 0.979. The K-means advantage over random grouping on $\mathrm{T}-\mathrm{AUC}_{\Delta}$ is p=0.016 (two-sided paired t-test, unadjusted, n=20, difference 0.080, 95% CI $[0.023, 0.138]$). The raw-window advantage over K-means is 0.067 (two-sided paired t-test, unadjusted, p=0.03, 95% CI $[0.013, 0.123]$), replicated on an untouched seed block (0.053, 95% CI $[0.025, 0.081]$, n=50).

Method	Pooled AUROC	Within-edge AUROC
Window mean	0.795	0.500
Window max	0.791	0.500
Bagged mean	0.792	0.500
Raw window (no grouping)	0.731	0.716
Random balanced	0.773	0.570
Gated hard	0.776	0.649
GMM weighted + argmax	0.776	0.608
GMM continuous soft	0.776	0.608
Temporal DP	0.765	0.562
Oracle label	0.806	0.814
Static ceiling $\mathrm{AUC}^{*}_{\mathrm{static}}$	0.979	

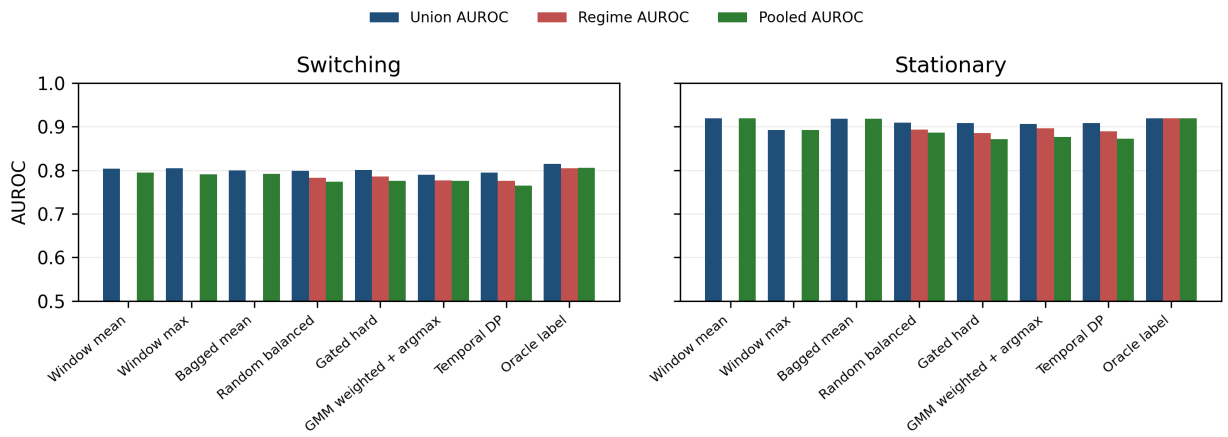


Figure S6: Union, regime and pooled AUROC for identical window scores. Union and regime are flat across aggregation rules; on the pooled target the static predictors and fitted clustering are close, and only the oracle assignment improves it.

Precision–recall and prevalence. Table S18 reports the two layers with prevalence, AUPRC lift, and normalized AP. With prevalence 0.249 for union edges and 0.174 for regime edges, the lift says the regime layer is easier (3.38× against 2.79×); normalized AP says it is harder (0.500 against 0.593). Both are correct, and they differ because lift has an upper bound of $1/\pi$ that grows as prevalence falls. We therefore do not claim an ordering: under the evaluated generator and protocol, union and regime AUROC are close, while their precision–recall behaviour differs with prevalence and with the choice of normalization.

Table S18: The two layers on the switching setting. Lift and normalized AP disagree about which layer is easier; we report both and draw no ordering.

Layer	Prevalence	AUROC	AUPRC	AUPRC / prev.	Normalized AP
Union	0.249	0.801	0.694	2.79×	0.593
Regime-specific	0.174	0.786	0.587	3.38×	0.500

Label error. Figure S7 sweeps symmetric label error on the K-means partition and, separately, on the true labels. Under K-means corruption union AUROC moves from 0.801 to 0.798, regime AUPRC from 0.587 to 0.568, and ARI from 0.292 to −0.041, while the pooled AUROC is almost flat (0.776 to 0.775) because the two fitted state scores are already similar. Under oracle corruption union AUROC moves from 0.814 to 0.794, regime AUPRC from 0.639 to 0.556, and the pooled AUROC from 0.806 to 0.775. In this configuration, pooled AUROC changes more under corruption of the true labels than of the fitted partition.

S6.3 Exploratory duration and record-length scans

We re-evaluate all 35 labelled entries of the existing mechanism grid at seeds 0–19, reusing the four repeated configurations: 31 distinct configurations give 620 configuration–seed fits. This is an exploratory extension on existing simulation seeds. Each method consumes the same local scores, and the paired quantity is

$$\Delta_{\text{temporal}} = \text{T}-\text{AUC}_{\Delta,\text{aggregation}} - \text{T}-\text{AUC}_{\Delta,\text{raw}}. \tag{S13}$$

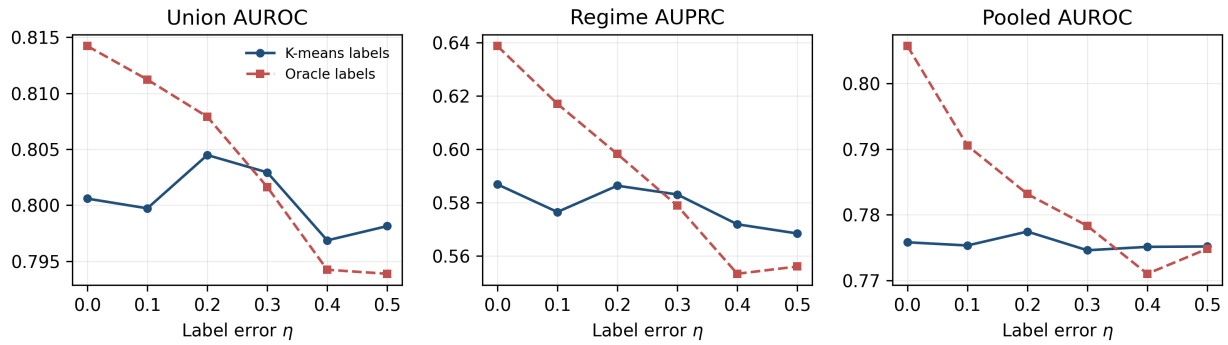


Figure S7: Sensitivity to label corruption applied to the fitted partition (solid) and the true labels (dashed). Union, regime-specific and pooled recovery respond differently to label corruption. Changes in pooled AUROC alone do not identify changes in within-edge temporal discrimination.

Table S19 and Figure S8 show the existing one-factor duration and length scans. In the former $h = 300$, $T = 1200$ and $M = 7$ are fixed while D varies; in the latter $h = D = 300$ and stride 150 are fixed while $T = 600, 1200, 3000$ gives $M = 3, 7, 19$. Thus M changes together with record length and number of switches; these are not independent interventions on M or a crossed h/D-by-M grid. Intervals are pointwise 95% percentile bootstrap intervals of paired seed differences (10,000 resamples), with no simultaneous-coverage or confirmatory significance claim. Undefined changing-edge targets are retained as missing, with evaluable denominators in the full CSV.

Duration changes the sign of the observed gain. The hard gate accepts 20/20 fits at every duration level. At $h/D = 1/2$, raw and hard within-edge temporal AUROC are 0.767 and 0.798, a paired gain of $+0.031$ with interval $[0.005, 0.057]$. At $h/D = 1/3$ the gain is $+0.017$ $[-0.013, 0.043]$; at $h/D = 1$ it is -0.067 $[-0.123, -0.013]$. At the full-cycle window $h/D = 2$, raw is already near chance (0.517) and hard aggregation changes it by -0.012 $[-0.021, -0.003]$. The small loss there does not represent good within-edge temporal discrimination. Oracle labels have positive mean gains across these levels, but at $h/D = 2$ their $+0.029$ interval $[-0.009, 0.064]$ includes zero. The earlier exploratory reference configuration's loss therefore does not extend uniformly across durations.

Matched random grouping at the selected positive cell. At $D = 600$, we enumerate every two-group partition with the fitted group sizes for each seed: 21 or 35 unlabelled partitions, 616 in total. Each partition rebuilds its own within-group means and is scored before its AUROC is averaged within the seed. The twenty simulation seeds, not the enumerated partitions, are the statistical units. The matched-random mean is 0.577, versus hard 0.798 and raw 0.767. Hard exceeds matched random by $+0.222$ $[0.193, 0.250]$, while matched random is below raw by -0.191 $[-0.204, -0.179]$ (Table S21). Thus hard's positive comparison with raw is accompanied by added value over this size-matched random control. It is not explained by arbitrary grouping-and-averaging under that control, but remains an exploratory result at a selected cell on existing seeds.

Connecting conditional gains to the two-state identity. Table S20 reports three hard-candidate conditions. Extending D from 300 to 600 at fixed $h = 300, M = 7$ raises both mean state-assignment discrimination and mean contrast concordance; the mean product increases from 0.298 to 0.596, recovering the hard AUROCs through Eq. (S3). The forced $M = 19$ candidate

has core-like state-assignment discrimination, a higher contrast concordance and mean product 0.359, giving hard AUROC 0.679 against raw 0.722. These profiles connect the conditions to the same identity without assigning independent causal roles to the two jointly estimated factors. The factors' marginal means are never multiplied to reconstruct a mean AUROC.

Gate fallback and executed aggregation answer different questions. For $M = 3, 7, 19$, gated hard gains are -0.017, -0.067 and -0.202; acceptance is 20/20, 20/20 and 2/20. At $M = 19$, forcing the same hard candidate on all seeds gives -0.042 $[-0.089, 0.006]$, whereas the gated pipeline gives -0.202 $[-0.229, -0.164]$. The 18 rejected fits emit a constant window-maximum graph with within-edge temporal AUROC 0.5; fallback therefore adds -0.160 to the all-seed mean difference relative to the forced candidates. Among the two fits that actually aggregate, hard gain is -0.046 against their own raw baseline (0.742). This selected subgroup is too small for an interval under our rule of at least five evaluable seeds. Section S6.4 reports accepted-subgroup contrasts for all methods on the same seeds, as well as gated hard execution counts. GMM and oracle have no acceptance gate and aggregate in all 20 fits per displayed cell; the shared hard-gate subgroup does not introduce a gate for either comparator.

The tested GMM does not provide a diffuse soft control. We distinguish responsibility-weighted state maps with argmax emission from continuous emission $\widehat{S}_w = \sum_k r_{wk} \widehat{G}_k$, where $\widehat{G}_k = \sum_v r_{vk} S_v / \sum_v r_{vk}$. Both use the same two-component diagonal GMM, three initialisations and simulation seed. Table S17 includes both: at the core point they each reach 0.608, giving continuous soft gain -0.109 $[-0.156, -0.060]$. Continuous soft mean gains are negative at all six distinct displayed configurations. All 620 GMM fits converge, but no window has maximum responsibility below 0.99; the largest mean normalised responsibility entropy is only 2.15×10^{-7}. Thus this fitted model is nearly a hard assignment and does not isolate the benefit of diffuse memberships. A post-run 12-decimal score-rounding check preserves this conclusion; it removes some tiny continuous/argmax ranking differences (Section S6.4). These results do not establish that soft aggregation generally fails.

Table S19: Paired temporal gain over raw windows on the existing duration and length scans (twenty seeds per cell). Raw is absolute within-edge temporal AUROC; all other score columns are Δ_{temporal}. Gate counts concern gated hard. Forced hard bypasses only that gate. GMM uses continuous responsibility emission; oracle averages by true window labels, without refitting. The reference cell appears once in each scan and is not an independent replication.

Scan/level	M	Gate	Raw	Gated Δ	Forced hard Δ	GMM soft Δ	Oracle Δ
Duration scan ($M = 7$)							
$h/D = 1/3$	7	20/20	0.792	+0.017	+0.017	-0.046	+0.060
$h/D = 1/2$	7	20/20	0.767	+0.031	+0.031	-0.061	+0.071
$h/D = 1$	7	20/20	0.716	-0.067	-0.067	-0.109	+0.097
$h/D = 2$	7	20/20	0.517	-0.012	-0.012	-0.019	+0.029
Length scan ($h/D = 1$)							
$M = 3$	3	20/20	0.684	-0.017	-0.017	-0.030	+0.004
$M = 7$	7	20/20	0.716	-0.067	-0.067	-0.109	+0.097
$M = 19$	19	2/20	0.722	-0.202	-0.042	-0.102	+0.160

Table S20: Three-condition two-state diagnostics, twenty seeds each at $h = 300$. Hard and Raw are within-edge temporal AUROCs. The identity uses the mean per-seed product, not the product of marginal means. The $M = 19$ row bypasses the gate and evaluates the unchanged hard candidate on all seeds. Factors are aligned by the same truth-overlap label matching used in the core diagnostic.

D	M	Hard output	$\mathbb{E}[J]$	$\mathbb{E}[C_{\text{edge}}]$	$\mathbb{E}[JC_{\text{edge}}]$	Hard	Raw
300	7	Gated	0.533	0.414	0.298	0.649	0.716
600	7	Gated	0.867	0.686	0.596	0.798	0.767
300	19	Forced	0.537	0.581	0.359	0.679	0.722

For the same seeds, writing T for within-edge temporal AUROC, the existing two-state identity gives

$$\overline{\Delta}_{\text{temporal}} = \overline{T}_{\text{hard}} - \overline{T}_{\text{raw}} = \frac{1}{2}\left[\overline{JC_{\text{edge}}} - \left(2\overline{T}_{\text{raw}} - 1\right)\right]. \tag{S14}$$

Hard aggregation improves on raw when its joint grouping–contrast ability $\overline{JC_{\text{edge}}}$ exceeds the temporal ability already present in raw scores, $2\overline{T}_{\text{raw}} - 1$. From $D = 300$ to 600 at $M = 7$, hard gains 0.149 while raw gains 0.051, so hard-minus-raw improves by 0.098; this is a descriptive rewriting of the existing results, not a new theorem or a general rule for choosing D.

S6.4 Matched partitions, gate subgroups, and GMM sensitivity

Exact matched grouping control. The selected positive cell has $D = 600, h = 300, T = 1200, M = 7$. For each fitted partition, use the smaller group size k and enumerate all $\binom{7}{k}$ memberships, with complement labels representing the same unlabelled partition. Observed sizes give 21 or 35 partitions per seed. Each partition estimates its own maps, emits them using its own assignments and yields one within-edge temporal AUROC. Average these AUROCs within seed before computing the twenty paired contrasts. Averaging prediction tensors first would be a different estimator and is not used. The corresponding partition rows and within-seed summaries are retained in the supplementary archive; the three contrasts are exploratory and their intervals are pointwise rather than a new confirmatory family.

Table S21: Selected-cell exact group-size-matched random control. Intervals resample the twenty seeds, after averaging partition AUROCs within each seed. The 616 partitions are not independent replications.

Comparison	Mean difference	Pointwise 95% CI	Seeds
Gated hard − raw	+0.031	[+0.005, +0.057]	20
Gated hard − matched random	+0.222	[+0.193, +0.250]	20
Matched random − raw	-0.191	[-0.204, -0.179]	20

The full grid contributes $35 \times 20 \times 5 = 3{,}500$ method rows, including the GMM argmax variant. Four duplicate configurations reuse their original observations, giving 620 distinct configuration–seed fits. For 240 of the 700 labelled configuration–seed contexts, no changing edge flips in the evaluation windows (including support-preserving controls and the rare-state condition); their within-edge temporal AUROC and gains remain undefined, rather than 0.5. Table S24 gives the evaluable counts and missing-target reasons for every labelled configuration. Repeated configuration IDs reuse the same observations rather than contributing independent data.

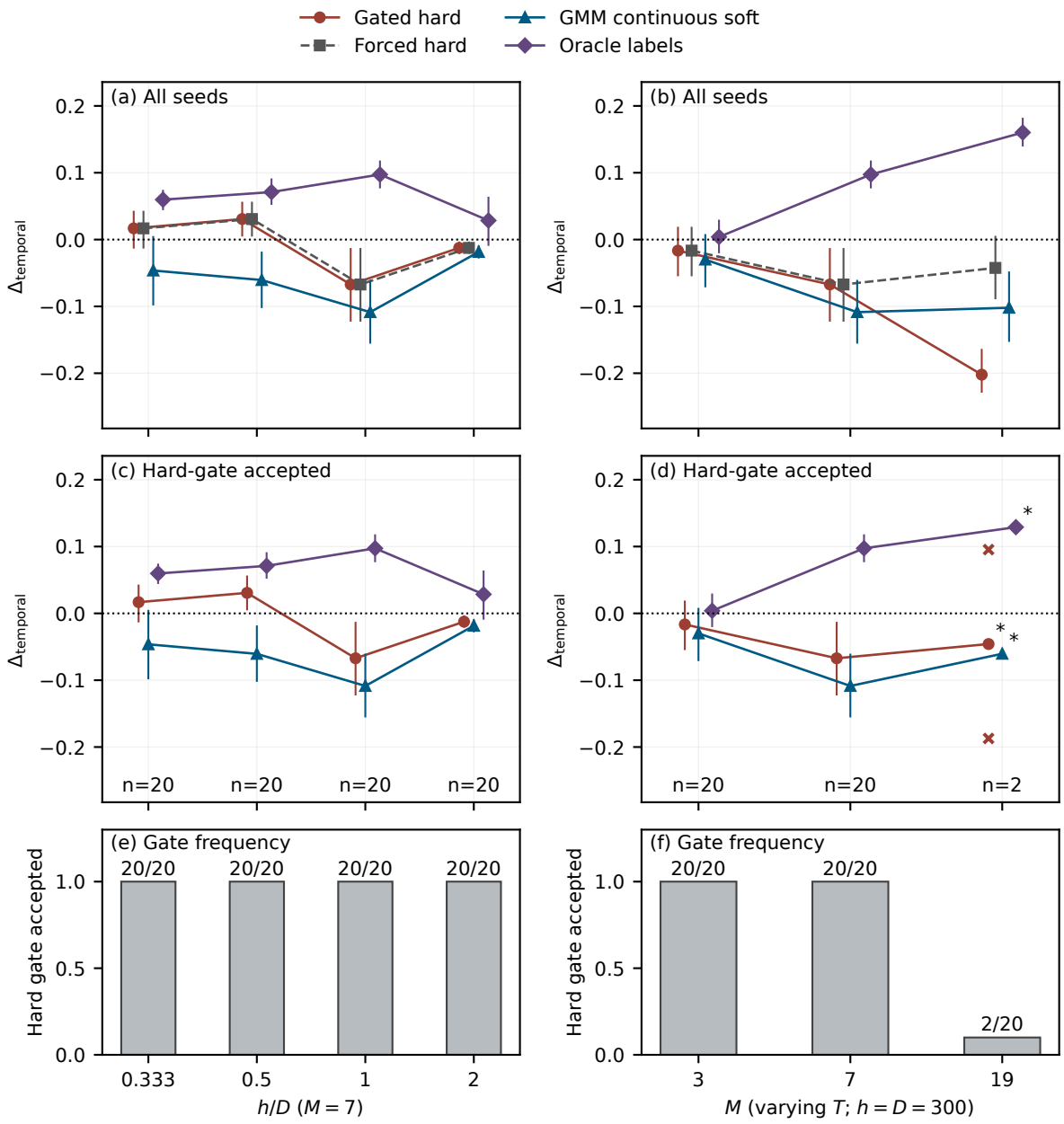


Figure S8: Temporal gain and gate selection at discrete existing grid levels. (a,b) All-seed paired gains and pointwise 95% bootstrap intervals; zero is the raw-window reference. (c,d) The same comparison on the common hard-gate accepted seeds; forced hard coincides with gated hard there and is omitted. Asterisks mean fewer than five seeds, so no interval is shown; crosses show the two individual hard gains at $M = 19$. These selected-subgroup means are descriptive. (e,f) Hard-gate acceptance; GMM and oracle execute without this gate in all twenty fits per cell. Left: vary D at fixed $M = 7$. Right: vary T to change M at fixed $h = D = 300$.

The summary CSV includes all-seed, common hard-gate accepted/rejected, and each method's actual execution subset. Table S22 shows the common accepted subset so each contrast retains its own paired raw baseline. For gated hard it is exactly the execution subset; forced hard is evaluated on all seeds. Ungated GMM and oracle also execute on all seeds. At $M = 19$, selection leaves two hard gains of opposite signs, so neither the accepted-only mean nor the all-seed forced mean supports a precise general conclusion about long-record aggregation.

Table S22: Paired gains on the common hard-gate accepted subset. n is the number of evaluable accepted seeds; Raw is their own absolute within-edge temporal AUROC. Intervals are pointwise paired-seed bootstrap intervals; [–] means fewer than five seeds, not a missing mean. GMM and oracle remain ungated comparators.

Scan/level	n	Raw	Gated Δ [95% CI]	GMM soft Δ [95% CI]	Oracle Δ [95% CI]
Duration scan ($M = 7$)					
$h/D = 1/3$	20	0.792	+0.017 [-0.013, +0.043]	-0.046 [-0.099, +0.005]	+0.060 [+0.044, +0.074]
$h/D = 1/2$	20	0.767	+0.031 [+0.005, +0.057]	-0.061 [-0.102, -0.018]	+0.071 [+0.052, +0.092]
$h/D = 1$	20	0.716	-0.067 [-0.123, -0.013]	-0.109 [-0.156, -0.060]	+0.097 [+0.077, +0.118]
$h/D = 2$	20	0.517	-0.012 [-0.021, -0.003]	-0.019 [-0.028, -0.009]	+0.029 [-0.009, +0.064]
Length scan ($h/D = 1$)					
$M = 3$	20	0.684	-0.017 [-0.055, +0.019]	-0.030 [-0.071, +0.008]	+0.004 [-0.020, +0.030]
$M = 7$	20	0.716	-0.067 [-0.123, -0.013]	-0.109 [-0.156, -0.060]	+0.097 [+0.077, +0.118]
$M = 19$	2	0.742	-0.046 [–]	-0.060 [–]	+0.129 [–]

The fitted diagonal GMM has almost binary responsibilities on this high-dimensional, few-window representation: every maximum responsibility exceeds 0.99 across all 620 fits, with no convergence warnings. This numerical behaviour limits what this comparator says about soft grouping. Table S23 reports a post-run sensitivity check on the six distinct focus cells, rounding both GMM emissions and raw scores to 12 decimal places without changing the primary method. Raw and argmax within-edge temporal AUROCs are unchanged for every checked seed. Some continuous-emission differences disappear; at $M = 19$ the continuous mean changes from 0.619735 to 0.619162, while its argmax counterpart is 0.615555. The largest per-seed continuous AUROC change is 0.01434. Mean continuous-minus-raw gains remain negative in all six cells after rounding. Tiny rank differences therefore receive no interpretation as evidence for a diffuse-membership benefit; primary continuous values are retained transparently.

Table S23: GMM emission and numerical sensitivity: absolute within-edge temporal AUROC averaged over the same twenty seeds. Responsibility-weighted state maps are shared by both GMM variants. Only the continuous emission is rounded in the last column; rounding raw and argmax also leaves their values unchanged.

Cell	h/D	M	GMM argmax	GMM continuous	Continuous, round 12
Reference	1	7	0.607690	0.607690	0.607690
$D = 150$	2	7	0.498190	0.498332	0.498190
$D = 600$	0.5	7	0.706884	0.706884	0.706884
$D = 900$	0.333	7	0.745898	0.745994	0.745898
$T = 600$	1	3	0.653904	0.653904	0.653904
$T = 3000$	1	19	0.615555	0.619735	0.619162

Table S24: Temporal-target coverage of the existing grid, twenty seeds per row. No change: no differing nonself support edge. No flip: differing support exists but its labels do not flip in the sampled evaluation windows. Both yield undefined within-edge temporal AUROC, not 0.5. Entries marked (=Cxx) reuse that earlier configuration's fits. Counts describe labelled contexts, not additional seeds.

ID	Configuration	M	Valid/20	No change	No flip
C00	Reference	7	20	0	0
C01	Overlap 0.0	7	20	0	0
C02	Overlap 0.25	7	20	0	0
C03	Overlap 0.75	7	20	0	0
C04	Overlap 1.0	7	0	20	0
C05	Duration 150	7	20	0	0
C06	Duration 600	7	20	0	0
C07	Duration 900	7	20	0	0
C08	Obs. noise 0.1	7	20	0	0
C09	Obs. noise 0.2	7	20	0	0
C10	Obs. noise 0.4	7	20	0	0
C11	Edge probability 0.1	7	20	0	0
C12	Edge probability 0.3	7	20	0	0
C13	Length 600	3	20	0	0
C14	Length 3000	19	20	0	0
C15	Effect scale 0.6	7	20	0	0
C16	Effect scale 1.4	7	20	0	0
C17	Maximum lag 5	7	20	0	0
C18	State shares 0.8/0.2	7	20	0	0
C19	State shares 0.95/0.05	7	0	0	20
C20	Duration 150 / Overlap 0.0	7	20	0	0
C21	Duration 150 / Overlap 1.0	7	0	20	0
C22	Duration 300 / Overlap 0.0 (=C01)	7	20	0	0
C23	Duration 300 / Overlap 1.0 (=C04)	7	0	20	0
C24	Obs. noise 0.0 / Overlap 0.0 (=C01)	7	20	0	0
C25	Obs. noise 0.0 / Overlap 1.0 (=C04)	7	0	20	0
C26	Obs. noise 0.4 / Overlap 0.0	7	20	0	0
C27	Obs. noise 0.4 / Overlap 1.0	7	0	20	0
C28	Control same support	7	0	20	0
C29	Control variance only	7	0	20	0
C30	Control mean only	7	0	20	0
C31	Control trend	7	0	20	0
C32	Control sign flip	7	0	20	0
C33	Control lag shift	7	20	0	0
C34	Control stationary	7	0	20	0

S6.5 Window-mixture and conditional denoising probes

An independent switched regression (two exogenous drivers predicting one target, coefficients alternating between $(0.9, 0)$ and $(0, 0.9)$ every D=150 steps, $T = 2400$, stride 75, OLS per window, then clustering the absolute coefficients; 100 paired seeds) reproduces a "vanish–recover–vanish" pattern: mean state ARI is 1.000 at h=75, 0.244 at h=150, −0.020 at h=300, 0.243 at h=450, and −0.028 at h=600. Windows of 300 and 600 make the population contrast vanish exactly ($q_{w,1} \equiv 0.5$, i.e. h=P and h=$2P$), while 450 restores partial contrast. The right question is therefore the *interaction* between window length and switching period (a rectangular average has frequency nulls that suppress certain periodic changes; the same sliding-window filtering issue is discussed for dynamic functional connectivity [Leonardi and Van De Ville, 2015]), not a monotone "shorter windows are better".

The earlier seven-window reference configuration is not at this mixing null. Its generator sets "switch_period = 300", i.e. $D = 300$ and $P = 600$, so a window of $h = 300$ is $h = D$, half a cycle. With the 100-step burn-in shifting the switch points off the window grid, the minority-state fraction across the seven windows ranges from 0.16 to 0.34, never the 0.5 the null requires, so β^*_w still varies across windows. We verify this on the real pipeline by freezing one coefficient tensor and schedule and re-simulating only the innovation noise (50 replicas): the population contrast — the standard deviation of the *signed population regression coefficients* β^*_w across windows, averaged over changing edges — is 0.038, versus 0.001 under a D=150 schedule (h=$2D$, the true null), while the per-replica scatter of the same signed OLS coefficients is 0.060. On this fixed mechanism and this summary scale the estimation scatter therefore exceeds the window contrast, consistent with a noise-limited reading. This is a single-instance conditional study, not a claim across mechanisms, and the two quantities live on the same signed coefficient scale (the cross-window standardised absolute scores that feed clustering are a different, nonlinear scale, so the 0.038/0.060 comparison is diagnostic rather than a direct clustering signal-to-noise ratio). A denoising intervention tests the reading: averaging the window Lasso scores over B independent noise replicas before clustering, holding the mechanism, window and scorer fixed, lifts the within-edge temporal AUROC from 0.792 at B=1 to 1.0 at B=16, and the state-contrast score C_{edge} (Section S3.2) from 0.583 to 1.0. Crucially, the state-assignment discrimination J is already 1.0 at B=1: the intervention repairs the per-state edge maps, not the state assignment. Lowering the estimation noise therefore recovers the within-edge temporal discrimination the single replica loses by sharpening the state-contrast score, so for this fixed mechanism the binding constraint is the state-map estimate, not state identification. This is a mechanism diagnostic, not a deployment recommendation (a real task does not observe multiple independent replicas of one mechanism), and it does not imply the twenty-seed exploratory panel would reach 1.0 under denoising. Two implementation details matter for this reading: the effective regression uses 297 rows, not 300 (the lagged design drops the first L=3 rows of each window), and the scores are locally z-scored before Lasso, so the design second moments differ from the raw VAR ones.

S6.6 Partition and refit configuration controls

The 2×2 factorial. Table S25 crosses partition (K-means, PELT, none) with re-estimation (window average, refit). Re-fitting lowers union AUROC at this resolution: K-means goes from 0.831 to 0.813 in the four-scenario mean and PELT from 0.829 to 0.793. PELT returns one segment per window in all 160 fits that cross its partition with either scoring rule (eighty partitions across four settings), so its window-average cell is numerically the window maximum and its refit cell is a per-window refit. A global VAR-Lasso is competitive: 0.832 averaged across the four scenarios,

better on the stationary (0.954 versus 0.908) and state-specific (0.801 versus 0.768) settings, worse when the states share a support (0.768 versus 0.847).

Table S25: 2×2 factorial over partition and re-estimation, union AUROC (twenty seeds per setting). "None" is a global VAR-Lasso with no partition.

Partition	Refit	Switching	State-specific	Shared	Stationary	Mean across scenarios
None	No	0.807	0.801	0.768	0.954	0.832
Returned hard groups	No	0.801	0.768	0.847	0.908	0.831
PELT (earlier grid)	No	0.804	0.771	0.847	0.893	0.829
Returned hard groups	Yes	0.784	0.768	0.783	0.915	0.813
PELT (earlier grid)	Yes	0.773	0.743	0.819	0.839	0.793
Oracle	Yes					0.862

A baseline-sensitivity study. Table S26 and Figure S9 compare four two-group/two-segment partitions with an unconstrained PELT reference. On the switching setting K-means aligns with the true states (ARI 0.292), while the best two contiguous segments (−0.061), the raw-VAR split (0.045) and the first-true-changepoint split (0.103) do not. These two configurations fail for different reasons: a fixed two-segment model lacks the capacity to represent the recurring $A \to B \to A \to B$ sequence, whereas PELT (which returns one segment per window, 7/7 in every fit) over-segments and re-fits very short ranges. Neither is a fully specified segmentation estimator, so this is a sensitivity study of segment count and refit configuration rather than a like-for-like method ranking. Union AUROC is nevertheless within 0.031 across all five partitions (0.773–0.804). The PELT penalty scan in Section S6.7 shows the realised count is set by the minimum segment length, not the penalty factor.

Table S26: Two-group/two-segment partitions and an unconstrained PELT reference (twenty seeds). ARI is against the true window regimes; U-AUROC is union AUROC after window averaging.

	Switching		State-specific	
Partition	ARI	U-AUROC	ARI	U-AUROC
K-means (2 groups)	+0.292	0.801	+0.471	0.768
Constrained 2 segments	-0.061	0.795	-0.042	0.755
Raw VAR split	+0.045	0.786	+0.042	0.751
PELT (earlier grid)	+0.000	0.804	+0.000	0.771
First-true-changepoint	+0.103	0.773	+0.103	0.744

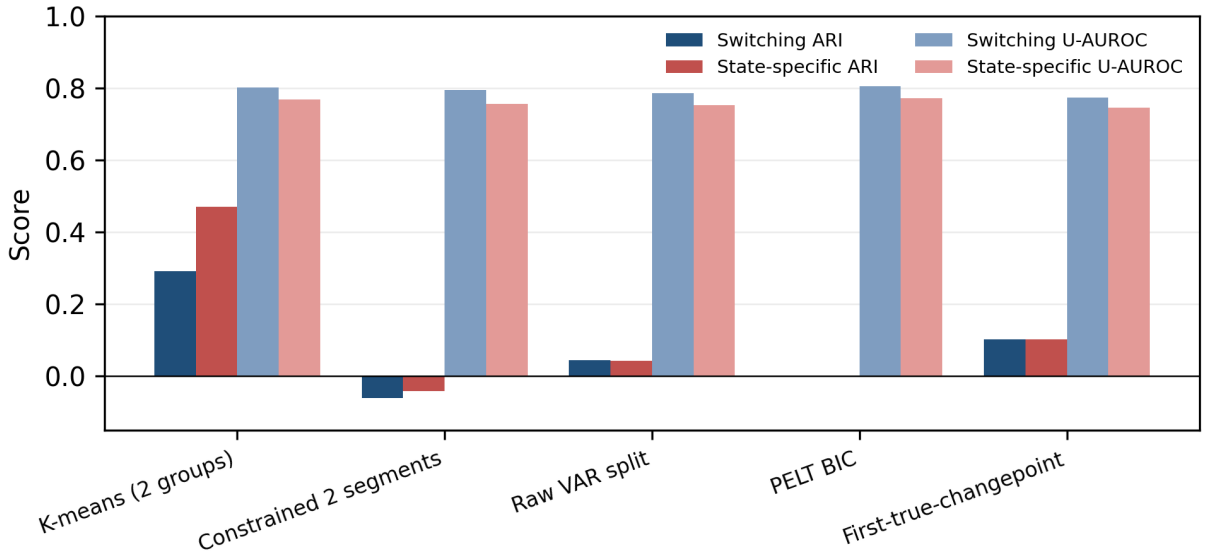


Figure S9: State alignment and union recovery for five partitions. A fixed two-segment model, even split at the first true change point, cannot represent recurring regimes; union recovery is nearly the same for all of them.

S6.7 PELT penalty and minimum-length scan

Table S27 varies the penalty multiplier from 0.25 to 4 and the minimum segment length from one to three windows in the earlier seven-window switching-core configuration, using the same twenty seeds per cell. The mean segment counts remain 7, 3 and 2 for minimum lengths one, two and three, respectively, across all tested multipliers, so the count is controlled by the minimum-length constraint here.

Table S27: PELT on the earlier seven-window switching reference: segment counts are unchanged across the tested penalty multipliers and depend on the minimum segment length.

Penalty factor	Min windows/segment	Mean segments	Union AUROC
0.25	1	7.0	0.804
0.25	2	3.0	0.799
0.25	3	2.0	0.791
0.50	1	7.0	0.804
0.50	2	3.0	0.799
0.50	3	2.0	0.791
1.00	1	7.0	0.804
1.00	2	3.0	0.799
1.00	3	2.0	0.791
2.00	1	7.0	0.804
2.00	2	3.0	0.799
2.00	3	2.0	0.791
4.00	1	7.0	0.804
4.00	2	3.0	0.799
4.00	3	2.0	0.791

S6.8 Auxiliary independent validation across four scenarios

This auxiliary independent validation uses fifty seeds 3000–3049 after fixing the method set and tests. Each seed contributes the mean over four scenarios before inference. Its primary family compares gated hard with static fallback with three configurations (window max, PELT+refit and constrained two-segment refit) on union AUROC, union AUPRC and pooled AUROC; Holm correction covers nine tests. A global VAR-Lasso is a descriptive reference with its own interval outside that family. This is a separate evidence block from Study A's paired-duration comparisons. Section S8.1 distinguishes it from development seeds.

Table S28 reports the result. gated hard with static fallback beats all three baselines on union AUROC and union AUPRC (AUROC gains of +0.0109, +0.0382 and +0.0234 against window max, PELT+refit and the constrained split; AUPRC gains of +0.0229, +0.0830 and +0.0398). On the pooled AUROC target the picture is different: gated hard with static fallback does not significantly beat the window maximum (+0.0021, Holm $p = 0.42$), and its pooled-AUROC gains (+0.042 against the constrained split, +0.102 against PELT+refit) are against two differently limited configurations: a fixed two-segment split that under-fits the recurring structure, and PELT, which over-segments into per-window ranges. Against the global VAR-Lasso, no union-AUROC advantage is detected under this protocol (+0.0006, 95% CI $[-0.0061, 0.0071]$), while union AUPRC is lower by 0.0206 ($[-0.0287, -0.0128]$). The pooled-AUROC difference is −0.0001. These comparisons do not establish equivalence. The pooled-AUROC advantage over the two segmentation configurations must be read alongside their distinct segment-count and refit limitations.

A separate within-edge temporal diagnostic on the switching-core scenario of this fifty-seed block gives hard-minus-raw −0.053, with paired 95% interval $[-0.081, -0.025]$: the absolute means are 0.660 for hard and 0.713 for raw. This diagnostic is not one of the nine primary cross-scenario tests.

Table S28: Auxiliary independent validation on fifty seeds (3000–3049), averaged over four scenarios within each seed. Δ is gated hard with static fallback minus the competitor; intervals are seed-clustered bootstrap 95%; Holm covers the nine primary tests.

Outcome	Comparison	Δ	95% CI	p_{Holm}	Sig.
Union AUROC	− window max	+0.0109	[+0.0073, +0.0146]	<0.0001	yes
Union AUROC	− PELT refit (earlier)	+0.0382	[+0.0317, +0.0448]	<0.0001	yes
Union AUROC	− constrained refit	+0.0234	[+0.0168, +0.0306]	<0.0001	yes
Union AUPRC	− window max	+0.0229	[+0.0169, +0.0289]	<0.0001	yes
Union AUPRC	− PELT refit (earlier)	+0.0830	[+0.0745, +0.0912]	<0.0001	yes
Union AUPRC	− constrained refit	+0.0398	[+0.0307, +0.0498]	<0.0001	yes
Pooled AUROC	− window max	+0.0021	[-0.0029, +0.0070]	0.4176	no
Pooled AUROC	− PELT refit (earlier)	+0.1015	[+0.0957, +0.1071]	<0.0001	yes
Pooled AUROC	− constrained refit	+0.0422	[+0.0370, +0.0473]	<0.0001	yes
Descriptive reference, not in the Holm family					
Union AUROC	− global VAR-Lasso	+0.0006	[-0.0061, +0.0071]	–	–
Union AUPRC	− global VAR-Lasso	-0.0206	[-0.0287, -0.0128]	–	–
Pooled AUROC	− global VAR-Lasso	-0.0001	[-0.0073, +0.0066]	–	–

S6.9 Fixed PELT plus graph-refit comparison on existing C inputs

This fixed comparison is fitted on the existing central Study C inputs; it adds an alternative segmentation/refit structure, not new independent simulation seeds.

PELT minimises a penalised contiguous-segment cost [Killick et al., 2012, ruptures]; it does not infer recurrent state identities. We apply the existing `ruptures` implementation to the same nonself window-score features at $h = 300$, stride 150, $M = 23$ in C. Features are centred and divided by their standard deviation plus 10^{-8}, then projected onto the first five principal components. The cost is squared Euclidean segment error, the penalty is fixed at $5\log(23)$, `min_size`=2 and `jump`=1. No true state or segment count is supplied, and no penalty is selected from results.

Detected window boundaries map to the midpoint between neighbouring integer window centres. The resulting half-open observed-time intervals partition $[0, T)$ without overlap. The original short-interval merge rule requires at least $\max(2L+5, L+3)$ samples. Each final interval is independently standardised and refit by VAR-Lasso with $L = 3$, $\alpha = 0.005$, the same self-lag regression design and nonself scoring convention. Window emissions use the final interval containing each centre, including after merging. Convergence warnings count as failed adapter cases; all failures, warnings, segment maps and actual ranges would remain saved without replacement.

The 60 prelisted C cases all succeed, with 644 segment refits and no warnings. At $D = 300$, 26 repeats choose 11 segments and four choose ten; at $D = 600$, 19 choose eleven, ten choose ten and one chooses nine. Thus 45 of 60 cases reach the minimum-length maximum of eleven segments. This behaviour is reported without retuning. The adapter changes the grouping/refit structure but still shares the original local linear features; it is not a reproduction of an entire published causal-discovery system.

Table S29: Segmentation/refit and hard-emission contrasts against the same raw windows in C, 30 independent paired repeats per duration. All intervals are ordinary pointwise 95% bootstrap intervals; no new hypothesis-test family is assigned. Standard AP and method-relative F1 are not interchangeable with the positive-change and raw-amplitude conventions. Repeated C forced-minus-raw within-edge temporal AUROC intervals use the original Study C summaries; other metrics retain comparison-specific streams.

D	Target	Forced hard minus raw	PELT + VAR minus raw	n
300	Pooled AUROC	$+0.106$ $[0.098, 0.114]$	$+0.013$ $[0.006, 0.020]$	30
300	Within-edge AUROC	-0.047 $[-0.084, -0.006]$	$+0.005$ $[-0.020, 0.027]$	30
300	AP_+	-0.295 $[-0.386, -0.198]$	$+0.038$ $[-0.057, 0.125]$	30
300	AP_{std}	-0.037 $[-0.084, 0.016]$	$+0.169$ $[0.117, 0.216]$	30
300	$F1_{raw}$	-0.225 $[-0.248, -0.201]$	$+0.053$ $[0.014, 0.088]$	30
300	$F1_{self}$	-0.036 $[-0.077, 0.004]$	$+0.065$ $[0.028, 0.097]$	30
300	Stable exceedance	-0.005 $[-0.009, -0.000]$	-0.000 $[-0.002, 0.002]$	30
600	Pooled AUROC	$+0.115$ $[0.106, 0.123]$	$+0.008$ $[0.001, 0.014]$	30
600	Within-edge AUROC	$+0.037$ $[0.002, 0.067]$	-0.006 $[-0.020, 0.008]$	30
600	AP_+	-0.055 $[-0.133, 0.026]$	-0.125 $[-0.205, -0.044]$	30
600	AP_{std}	$+0.040$ $[-0.020, 0.102]$	-0.011 $[-0.067, 0.048]$	30
600	$F1_{raw}$	-0.100 $[-0.121, -0.079]$	-0.008 $[-0.055, 0.040]$	30
600	$F1_{self}$	$+0.150$ $[0.093, 0.205]$	$+0.008$ $[-0.043, 0.060]$	30
600	Stable exceedance	-0.008 $[-0.011, -0.006]$	-0.001 $[-0.002, -0.000]$	30

S6.10 Matched-protocol computational cost

The gated hard policy, raw window maximum and PELT with window averaging each take approximately 0.051 s per fit; PELT with contiguous-range refitting takes 0.063 s. Table S30 reports these measurements on one CPU worker under the same timing protocol.

Table S30: Wall-clock cost per fit under the same protocol, one CPU worker.

Method	Seconds per fit	Source
Window max	0.051	measured
Gated hard	0.051	measured
PELT + refit	0.063	measured
PELT + window average	0.051	measured

S7 Industrial alignment and reference targets

S7.1 ESS subsystem alignment and imputation sensitivity

ESS. The ESS benchmark provides 227 process variables, an expert subsystem graph over 23 subsystems, and one hour of one-second data [Mogensen et al., 2024]. We z-score each variable, average the variables inside each subsystem, and keep the 23 subsystems in the expert matrix (45 variables outside those subsystems are dropped). We compare forward fill, backward fill, linear interpolation, and column means. Forward and backward fill use full-hour column means for remaining missing entries; standardisation also uses the full hour. These are offline analyses, not strictly online imputations. Lags are collapsed by maximum before comparison because the expert graph is rolled and not lag-specific. PELT returns 23 segments under each imputation, each refitted on a disjoint time range without cross-boundary lag rows. The refit mapping and its implementation correction are disclosed below. For each edge set and each null seed, 5,000 graphs target the uniform distribution over simple directed graphs with the same labelled in- and out-degrees, excluding self-loops. The switch chain uses symmetric two-edge destination exchanges (90%) and directed three-cycle reversals (10%); proposals sample edge indices or vertices independently with replacement, not from currently valid moves [LaMar, 2009]. Rejections retain the current graph; burn-in is $100|E|$ proposals and thinning $10|E|$. Section S7.2 specifies the proposals and reports acceptance and tail diagnostics. Samples are shared across methods and imputations. These are approximate finite-chain null samples, with upper-tail $p = (1 + b)/5001$ for b null AUROCs at least as large as observed. The plus-one formula does not make sequential MCMC samples exchangeable or guarantee an exact finite-sample test [Howes, 2026]. Null seeds 0–4 assess Monte Carlo variability on the same hour, with seed 0 retained as the reference. The forward-fill strategy remains the primary analysis: its eight tests (four methods by two edge sets) are Holm-corrected as one family within each seed. The other imputations are exploratory checks; a combined 32-test correction is also computed separately within each seed.

Table S31 reports the forward-fill result and the range over imputations. No method exceeds 0.58 AUROC against all expert edges in the forward-fill result: gated hard with static fallback reaches 0.571, PELT 0.565 and lagged correlation 0.554. On strong edges they reach 0.587, 0.555 and 0.677, respectively. At reference seed 0, lagged correlation on strong edges has raw $p = 0.0032$ and Holm-adjusted 0.0256. Its adjusted values span $[0.0160, 0.0368]$ and remain below 0.05 in all 5/5 seeds; the other seven forward-fill tests remain above 0.05 in every seed. The all-edge gated

hard with static fallback result has reference raw $p = 0.0156$ and adjusted 0.1092; PELT's strong-edge raw $p = 0.1278$ becomes 0.4047. Under the combined 32-test correction, strong-edge lagged correlation with linear imputation has reference adjusted 0.0448 and spans $[0.0192, 0.0512]$ (4/5 rejections); mean imputation rejects in only 1/5 seeds. No combined-family rejection is stable across all five seeds. All-edge rankings vary with imputation: gated hard with static fallback is first under forward fill, PELT under backward fill, and lagged correlation under linear and mean imputation. The absolute AUROC moves by up to 0.069 for gated hard with static fallback. These finite-chain degree-null results remain exploratory and do not establish causal graph recovery or within-edge temporal discrimination. A significant comparison to the null for one method and a nonsignificant comparison for another do not establish a significant difference between those methods.

Table S31: ESS subsystem alignment with forward-fill imputation. Imputation range spans AUROC over four strategies. Raw and Holm p use reference seed 0; Holm range spans seeds 0–4, each with 5,000 degree-preserving null graphs and its own eight-test forward-fill family.

Method	Expert edges	AUROC	Imputation range	Raw p	Holm p	Holm range
Gated hard	all edges	0.571	[0.528, 0.571]	0.0156	0.1092	[0.1092, 0.1316]
Gated hard	strong only	0.587	[0.518, 0.587]	0.0322	0.1932	[0.1728, 0.2208]
Window max	all edges	0.545	[0.504, 0.545]	0.2693	0.5051	[0.5051, 0.5179]
Window max	strong only	0.541	[0.518, 0.572]	0.2525	0.5051	[0.5051, 0.5179]
PELT + refit	all edges	0.565	[0.545, 0.565]	0.1012	0.4047	[0.3927, 0.4175]
PELT + refit	strong only	0.555	[0.535, 0.661]	0.1278	0.4047	[0.3927, 0.4175]
Lagged correlation	all edges	0.554	[0.554, 0.584]	0.0460	0.2300	[0.1776, 0.2300]
Lagged correlation	strong only	0.677	[0.656, 0.691]	0.0032	0.0256	[0.0160, 0.0368]

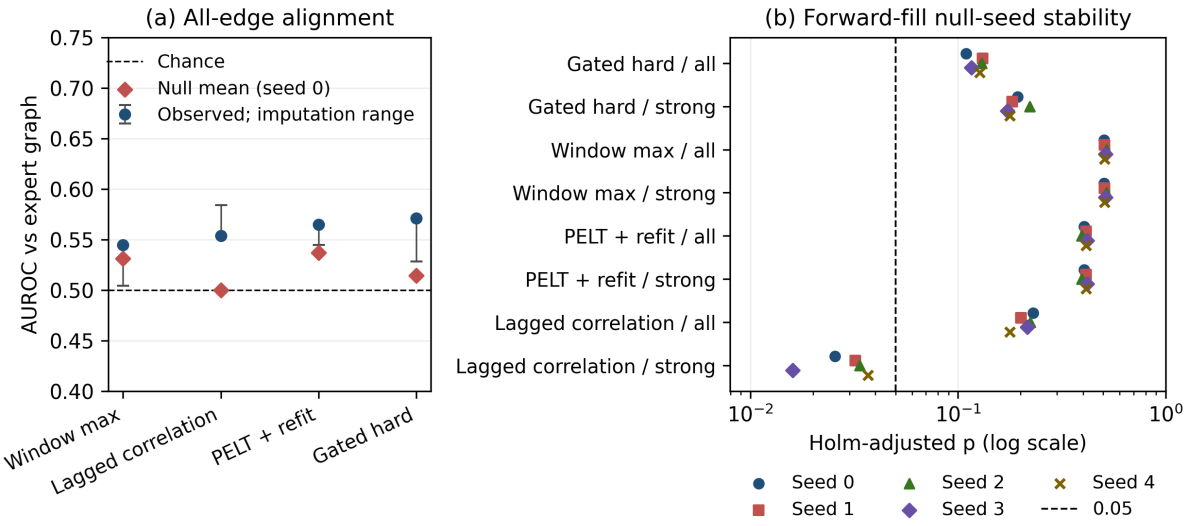


Figure S10: ESS alignment and null-seed stability. (a) Forward-fill all-edge AUROC: circles are observed scores, whiskers span four imputations (not a confidence interval), diamonds are degree-null means at reference seed 0, and the dashed line is chance. (b) Each marker gives one seed's forward-fill Holm-adjusted p from 5,000 null graphs; the dashed line marks 0.05.

ESS contiguous-range correction. The ESS adapter maps adjacent window centres to half-open time ranges, refits VAR-Lasso separately on each range without cross-boundary lag rows, and takes the maximum score across segments and lags. Its former merge test treated touching endpoints as overlap and collapsed all ranges. The correction retains the 23 ranges $[0, 224), [224, 374), \ldots, [3374, 3600)$. Direct global refitting reproduces the former observed AUROCs; the corrected 200- and 5,000-draw reports have identical observed AUROCs and AUPRCs. Thus the changed observed scores reflect the range-mapping repair, not increased null draws. The synthetic factorial, auxiliary validation and constrained segmentation tables use a separate refit implementation that already keeps touching runs separate. All 24 factorial rows at seed 0 replay exactly; a regression check also equates the two adapters on adequate chronological segments. Detailed implementation records remain in the archive.

S7.2 Degree-preserving null proposals and finite-chain diagnostics

Target and proposals. The target is uniform on all loop-free simple directed graphs on the same labelled vertices with the observed in- and out-degree at each vertex. Reciprocal edges are allowed; parallel edges are not. Starting at the observed expert graph, each attempted transition is:

1. With probability 0.9, draw an ordered pair of edge indices independently and uniformly with replacement. For edges $(a, b), (c, d)$, replace them by $(a, d), (c, b)$ only if neither is a loop or already present; otherwise stay.

2. With probability 0.1, draw an ordered vertex triple independently and uniformly with replacement. Stay if vertices repeat. If the triple has a directed three-cycle in either orientation and all three reverse edges are absent, reverse it; otherwise stay. No list of currently reversible cycles is sampled.

The reverse edge exchange has the same ordered-pair probability, since $|E|$ is fixed; each reversible triple has the same ordered-triple probability in both orientations. Thus the kernel is symmetric, so uniform measure is stationary without a Metropolis correction. The move types connect the fixed-degree realisation space [LaMar, 2009]; this gives no quantitative mixing-time guarantee for these ESS chains. Every invalid proposal counts as a self-transition.

Diagnostics and numerical uncertainty. Table S32 replays the same five 5,000-draw chains per edge set with counters that do not consume random numbers. All ten chains reproduce the stored lagged-correlation null means and exceedance counts. Figure S11 shows complete reference traces and sample autocorrelations for this fixed score. The tail indicator is $I_b = \mathbf{1}\{\mathrm{AUC}_b \geq \mathrm{AUC}_{\mathrm{obs}}\}$. We estimate its ESS by an initial positive, monotone sequence of paired sample autocorrelations, cap ESS at 5,000, and report $\widehat{\mathrm{MCSE}} = \sqrt{\widehat{\mathrm{Var}}(I_b)/\widehat{\mathrm{ESS}}}$. These single-chain estimates are descriptive, especially for rare events; the CSV additionally reports AUROC ESS and a 20-batch means MCSE. For the reference strong-edge chain, $b = 15$, tail ESS is estimated as 5,000 and MCSE as 0.00077 (batch MCSE 0.00070). Extending only seed 0 to $1{,}000|E|$ burn-in gives strong-edge raw $p = 0.0022$ instead of 0.0032. These checks reveal no marked retained-trace dependence for the reported statistic, but neither establish mixing nor remove finite-chain bias. Approximate p and Holm values remain exploratory; the plus-one adjustment alone provides no exact MCMC test guarantee [Howes, 2026].

Table S32: Lagged-correlation null-chain diagnostics. Acceptance rates are per attempted move of that type after burn-in, including rejected proposals in the denominator. b, lag-one autocorrelation ρ_1, ESS and MCSE concern the tail indicator. Rows with burn/$|E|$ = 1,000 are a seed-0 sensitivity check, not replacements for the primary protocol. All rows retain 5,000 draws.

Edges	Seed	Burn/$\lvert E\rvert$	2-edge (%)	3-cycle (%)	b	ρ_1	ESS	MCSE $\times 10^3$
All	0	100	62.94	0.238	229	−0.011	5000	2.957
All	1	100	62.88	0.260	200	+0.026	4749	2.844
All	2	100	62.90	0.240	222	+0.024	4499	3.071
All	3	100	62.94	0.238	214	+0.009	4888	2.895
All	4	100	62.94	0.234	168	−0.004	5000	2.549
All	0	1000	62.98	0.253	209	+0.011	4783	2.894
Strong	0	100	73.34	0.042	15	−0.003	5000	0.774
Strong	1	100	73.29	0.047	19	−0.004	5000	0.870
Strong	2	100	73.26	0.043	20	−0.004	5000	0.893
Strong	3	100	73.30	0.053	9	−0.002	5000	0.600
Strong	4	100	73.25	0.048	22	−0.004	4657	0.970
Strong	0	1000	73.26	0.039	10	−0.002	5000	0.632

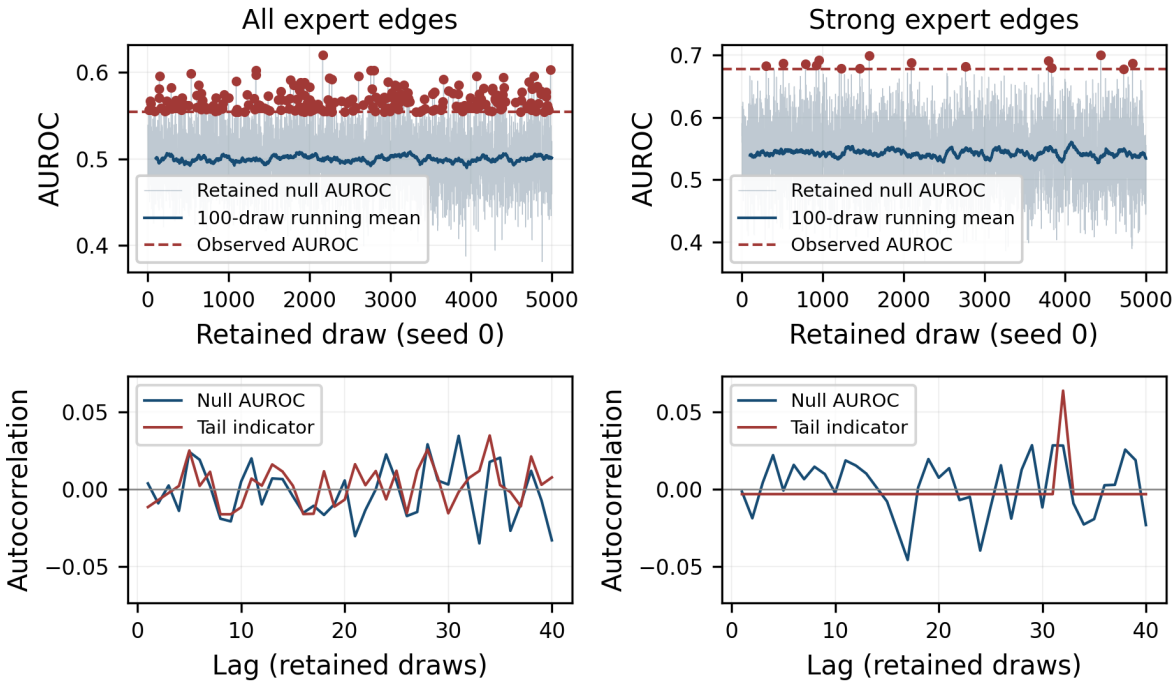


Figure S11: Reference ESS null traces (seed 0, burn-in 100$|E|$, thinning 10$|E|$). Top: all 5,000 retained AUROCs, a 100-draw running mean, and the observed threshold; red points mark exceedances. Bottom: sample autocorrelation of the AUROC and of its tail indicator at retained-draw lags 1–40. Small empirical autocorrelations are not a proof of adequate mixing.

S7.3 Operating-condition labels are not graph truth

On HydraulicSystems and AReM, condition labels mark changes in machine settings or activities, not changes in a stationary VAR. A change-point test calibrated on synthetic coefficient breaks accepts 34.2% of 193 Hydraulic condition boundaries against 30.0% of stable segments, and 100% of both in AReM, so those labels cannot be used as graph truth.

S8 Reproducibility and implementation notes

S8.1 Implementation scope and analysis records

Earlier selection and seed blocks. Seeds 2000–2049 were used while correcting the PELT adapter and were not treated as an independent validation block afterwards. The fixed earlier comparison was evaluated on seeds 3000–3049, with its own nine-test family and targets as described in Section S6.8. It was untouched by model selection but not a prospectively registered sample. Neither block supplies the new independent paired-duration replications of the main article.

Retained-output consistency. All 1,400 existing hard/oracle temporal measurements and gate decisions agree with the earlier grid, as do the 60 core raw/hard/oracle measurements and the 20 historical core GMM pooled values. The twenty-seed grid, numerical GMM sensitivity check and exact matched grouping analysis retain their separate output records. Replaying forty old fits to recover labels and state maps adds no seeds and does not turn the selected-cell comparison into an independent confirmation. The earlier responsibility-weighted GMM maps with argmax emission and the continuous-emission variant are reported separately.

The archived original implementation is identified in code as STCG-v1. Here its prediction rule is called gated hard, explicitly including the static window-maximum fallback. Forced hard and raw fallback are different emission policies with the same fitted candidate and gate decision.

S8.2 Post hoc diagnostics of saved generator outputs

The audit reads the saved final coefficient tensors and observed trajectories; it neither simulates missing observations nor refits a scorer. Table S33 counts each independent mechanism once for the radius range. Study A's 100 mechanisms generate 200 duration-specific paths shared by Lasso and OLS; Study B's 50 mechanisms generate 300 paths. All 500 saved observed paths are finite. The maximum absolute observed values are 0.532854 and 0.555313 in A and B. The endpoint ratio is the RMS of all nodes in the final 300 observed samples divided by the RMS in the first 300. Its ranges are $[0.930770, 1.050883]$ and $[0.936157, 1.063834]$. These are post hoc descriptive checks with no prespecified stability cutoff; the endpoint ratio is not a fitted growth rate and never controls inclusion.

Saved observations cover full-time intervals $[100, 1300)$ in A and $[100, 3700)$ in B. The original simulator checks finiteness before dropping burn-in; this independent audit examines only the saved post-burn-in series. It does not reconstruct the missing 100 burn-in observations. Pre-rescaling radii were not saved, so the number of applied rescalings is not inferred from final radii alone. Finite observed paths and state-wise radii do not establish general stability under arbitrary switching.

State A has no self support. State B self-edge counts span 0–7 across Study A mechanisms and 0–6 across Study B mechanisms. Realised shared/A support fractions span $[0.281, 0.742]$ and $[0.333, 0.706]$, rather than equalling 0.5 in every draw. The retained supports exclude activation of either empty-support safeguard in these 150 mechanisms. Full per-mechanism and per-path

diagnostics are in `results/information_sciences/generator_audit/`; its manifest binds the audit tool, outputs and unchanged frozen-source snapshots.

Table S33: Post hoc diagnostics of retained generator outputs. Radius ranges use final companion matrices; paths are paired within mechanism and are not independent mechanism replicates. RMS uses the first/last 300 saved observed samples and all nodes. No numerical cutoff or sample selection is introduced.

Study	Mechanisms	Paths	Actual radius range	Max $\lvert X\rvert$	Late/early RMS range	Finite paths
A	100	200	[0.381, 0.676]	0.533	[0.931, 1.051]	200/200
B	50	300	[0.404, 0.627]	0.555	[0.936, 1.064]	300/300

S8.3 Selected same-seed generation and fitting checks

We separately regenerate and fit A seeds 50000–50001 and C seeds 70000–70001 at both durations, eight selected pipeline cases. RNG inputs and trajectories are generated anew and saved before fitting; fitted arrays are not used to initialise the reruns. The isolated environment matches the 47 pinned dependencies. All 228 array comparisons, 132 original-metric comparisons and 132 recomputed-metric comparisons are exact on this machine, rather than merely within the predeclared 10^{-10} relative/10^{-12} absolute tolerance. These existing-seed reruns test the selected pipelines; they do not add independent seeds, test every setting from scratch or constitute an independent algorithm implementation.

The existing-output sensitivity analysis and the fixed segmentation/refit comparison have source/input/output hash closures and internal execution records, with no public-preregistration claim. The archive includes the fixed protocols, per-case arrays, summaries and evidence index. Its read-only entry point recomputes saved-output metrics and intervals without fitting; the display-rebuild route reconstructs figures and tables. New generation and fitting require an explicit new destination and are documented separately. The accompanying archive supplies the current presentation alongside the unchanged retained scientific inputs and fitted outputs. The corresponding public release is identified in Section S8.4.

S8.4 Access, cached replay, and rebuilding

The companion archive `information_sciences_reproducibility.zip` contains manuscript sources, configurations, retained inputs, emissions, assignments, truth, per-repeat results and presentation builders for this presentation revision. The complete archive is publicly available in version `v2026.10.28-layout.1` at https://github.com/xschang1002/latent-state-graph-recovery/releases/tag/v2026.10.28-layout.1. Its root README gives environment setup, data-access requirements and the execution index. The basic entries are:

```
python scripts/reproduce_paper.py --verify
python scripts/reproduce_paper.py --rebuild
```

Verification recomputes metrics, intervals and provenance from retained outputs without fitting or writing. Rebuild regenerates deterministic retained-output analyses and presentation files; it does not generate a new simulation sample. The separately documented simulation/fitting entries require frozen protocols and a new destination. Adding `--compile` uses an installed TeX engine. Public industrial datasets used by older auxiliary protocols may require separate acquisition, as distinguished from the packaged data in the README.

The machine-readable evidence index and research guide locate every study's protocol, per-case arrays, comparison families, missing-target coverage, source/output hashes and replay commands. Detailed historical script lists, development sequence and clock records remain in the repository rather than serving as evidence of new independent experiments. The earlier risk, occupancy, Brier, selection and pooled-refit probes are mechanism illustrations and are not pooled with the simulation-study tables. Internal source-lock times and filesystem observations are distinguished from externally attested registration; no public-preregistration claim is made.

References

Yiqun T. Chen and Daniela M. Witten. Selective inference for k-means clustering. *Journal of Machine Learning Research*, 24(152):1–41, 2023.

Stéphan Clémençon, Gábor Lugosi, and Nicolas Vayatis. Ranking and empirical minimization of U-statistics. *The Annals of Statistics*, 36(2):844–874, 2008. doi: 10.1214/009052607000000910.

Peter Hall, J. S. Marron, and Amnon Neeman. Geometric representation of high dimension, low sample size data. *Journal of the Royal Statistical Society: Series B*, 67(3):427–444, 2005. doi: 10.1111/j.1467-9868.2005.00510.x.

Michael Howes. Markov Chain Monte Carlo significance tests. *Statistical Science*, 41(1):188–200, 2026. doi: 10.1214/24-STS958. URL https://doi.org/10.1214/24-STS958.

Rebecca Killick, Paul Fearnhead, and Idris A. Eckley. Optimal detection of changepoints with a linear computational cost. *Journal of the American Statistical Association*, 107(500):1590–1598, 2012.

M. Drew LaMar. On uniform sampling simple directed graph realizations of degree sequences. arXiv:0912.3834, 2009. URL https://arxiv.org/abs/0912.3834.

Nora Leonardi and Dimitri Van De Ville. On spurious and real fluctuations of dynamic functional connectivity during rest. *NeuroImage*, 104:430–436, 2015. doi: 10.1016/j.neuroimage.2014.09.007.

Marco Loog, Jesse H. Krijthe, and Manuele Bicego. Also for k-means: More data does not imply better performance. *Machine Learning*, 112:3033–3050, 2023. doi: 10.1007/s10994-023-06361-6.

Søren Wengel Mogensen, Karin Rathsman, and Per Nilsson. Causal discovery in a complex industrial system: A time series benchmark. In *Conference on Causal Learning and Reasoning (CLeaR)*, volume 236 of *Proceedings of Machine Learning Research*, pages 1218–1236, 2024. URL https://proceedings.mlr.press/v236/mogensen24a.html.

Farimah Poursafaei, Shenyang Huang, Kellin Pelrine, and Reihaneh Rabbany. Towards better evaluation for dynamic link prediction. In *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*, 2022.

ruptures. Pelt: implementation and API reference. Official ruptures documentation. URL https://centre-borelli.github.io/ruptures-docs/code-reference/detection/pelt-reference/. Accessed 25 October 2026.

scikit-learn developers. average_precision_score. Official scikit-learn documentation. URL https://scikit-learn.org/stable/modules/generated/sklearn.metrics.average_precision_score.html. Accessed 25 October 2026.

Robert Tibshirani, Guenther Walther, and Trevor Hastie. Estimating the number of clusters in a data set via the gap statistic. *Journal of the Royal Statistical Society: Series B*, 63(2):411–423, 2001.